

# EEV SERIES WALL MOUNT AIR HANDLERS

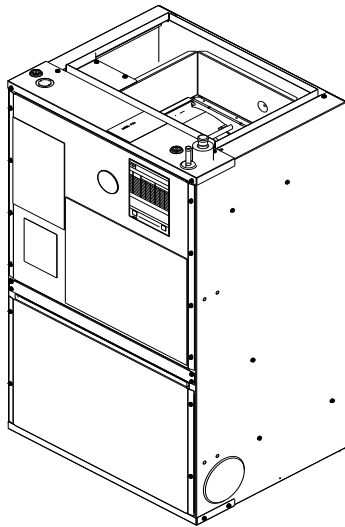
## INSTALLATION INSTRUCTIONS & OPERATING INSTRUCTIONS

© 2026 Daikin Comfort Technologies Manufacturing, Inc.  
19001 Kermier Rd., Waller, TX 77484

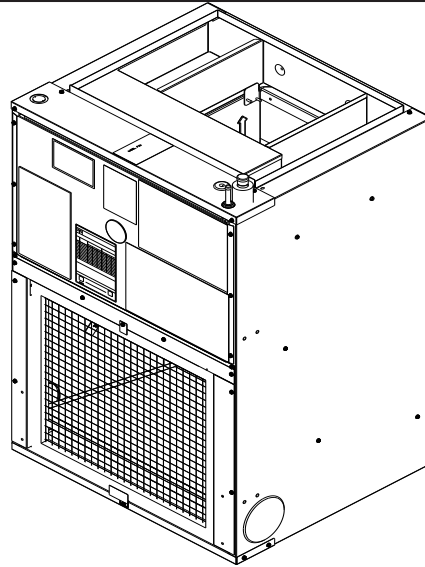
[www.daikincomfort.com](http://www.daikincomfort.com)

[www.goodmanmfg.com](http://www.goodmanmfg.com)

P/N: IOD-4069 Date: January 2026



AWVE 12,18,24



AWVE 30,36



### WARNING

ONLY PERSONNEL THAT HAVE BEEN TRAINED TO INSTALL, ADJUST, SERVICE, MAINTENANCE OR REPAIR (HEREINAFTER, "SERVICE") THE EQUIPMENT SPECIFIED IN THIS MANUAL SHOULD SERVICE THE EQUIPMENT.

THIS EQUIPMENT IS NOT INTENDED FOR USE BY PERSONS (INCLUDING CHILDREN) WITH REDUCED PHYSICAL, SENSORY OR MENTAL CAPACITIES, OR LACK OF EXPERIENCE AND KNOWLEDGE, UNLESS THEY HAVE BEEN GIVEN SUPERVISION OR INSTRUCTION CONCERNING USE OF THE APPLIANCE BY A PERSON RESPONSIBLE FOR THEIR SAFETY.

CHILDREN SHOULD BE SUPERVISED TO ENSURE THAT THEY DO NOT PLAY WITH THE EQUIPMENT.

THE MANUFACTURER WILL NOT BE RESPONSIBLE FOR ANY INJURY OR PROPERTY DAMAGE ARISING FROM IMPROPER SUPERVISION, SERVICE OR SERVICE PROCEDURES. IF YOU SERVICE THIS UNIT, YOU ASSUME RESPONSIBILITY FOR ANY INJURY OR PROPERTY DAMAGE WHICH MAY RESULT. IN ADDITION, IN JURISDICTIONS THAT REQUIRE ONE OR MORE LICENSES TO SERVICE THE EQUIPMENT SPECIFIED IN THIS MANUAL, ONLY LICENSED PERSONNEL SHOULD SERVICE THE EQUIPMENT. IMPROPER SUPERVISION, INSTALLATION, ADJUSTMENT, SERVICING, MAINTENANCE OR REPAIR OF THE EQUIPMENT SPECIFIED IN THIS MANUAL, OR ATTEMPTING TO INSTALL, ADJUST, SERVICE OR REPAIR THE EQUIPMENT SPECIFIED IN THIS MANUAL WITHOUT PROPER SUPERVISION OR TRAINING MAY RESULT IN PRODUCT DAMAGE, PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.

**QUEBEC DISCLOSURE REGARDING AVAILABILITY OF REPLACEMENT PARTS, REPAIR SERVICES AND INFORMATION FOR MAINTENANCE AND REPAIR:**  
**DAIKIN COMFORT TECHNOLOGIES MANUFACTURING, INC. (DAIKIN) DOES NOT GUARANTEE THE AVAILABILITY OF (1) REPLACEMENT PARTS; (2) REPAIR SERVICES; AND (3) INFORMATION NECESSARY TO MAINTAIN OR REPAIR PRODUCTS, WITHIN THE MEANING OF SECTION 39.1 OF THE CONSUMER PROTECTION ACT, CQLR, c P-40.1 AND SECTION 79.18 OF THE REGULATION RESPECTING THE APPLICATION OF THE CONSUMER PROTECTION ACT, CQLR, c P-40.1, R. 3.**

**AVIS POUR LE QUÉBEC CONCERNANT LA DISPONIBILITÉ DES PIÈCES DE RECHANGE, DES SERVICES DE RÉPARATION ET DES RENSEIGNEMENTS POUR L'ENTRETIEN ET LA RÉPARATION :** DAIKIN COMFORT TECHNOLOGIES MANUFACTURING, INC. (DAIKIN) NE GARANTIT PAS LA DISPONIBILITÉ (1) DES PIÈCES DE RECHANGE, (2) DES SERVICES DE RÉPARATION ET (3) DES RENSEIGNEMENTS POUR L'ENTRETIEN ET LA RÉPARATION, AU SENS DE L'ARTICLE 39.1 DE LA LOI SUR LA PROTECTION DU CONSOMMATEUR, RLRQ c P-40.1 ET DE L'ARTICLE 79.18 DU RÈGLEMENT D'APPLICATION DE LA LOI SUR LA PROTECTION DU CONSOMMATEUR, RLRQ c P-40.1, R 3.



### WARNING

DO NOT BYPASS SAFETY DEVICES.





## RECOGNIZE THIS SYMBOL AS A SAFETY PRECAUTION.

**NOTE:** SPECIFICATIONS AND PERFORMANCE DATA LISTED HEREIN ARE SUBJECT TO CHANGE WITHOUT NOTICE.

**NOTE:** NEVER OPERATE THE UNIT WITHOUT THE RECOMMENDED FILTER OR THE OUTER PANEL INSTALLED.

**NOTE:** THIS EQUIPMENT IS APPROVED TO USE ONLY WITH R-32 REFRIGERANT.

**NOTE:** IF AN "ED" ERROR IS ENCOUNTERED ON STARTUP, VERIFY THAT THE ELECTRIC HEATER DIP SWITCHES HAVE BEEN SET TO THE APPROPRIATE HEATER SIZE. SEE TABLES 7 AND 8 FOR THE HEATER KIT AIRFLOW DELIVERY AND DIP SWITCH SETTINGS.

**NOTE:** THE INSTALLER MUST REFER TO SECTION 13.7 FOR MAXIMUM POSSIBLE REFRIGERANT CHARGE AND MINIMUM (AIR CONDITIONED) ROOM AREA WHICH MUST BE RECORDED ON THE SERIAL PLATE.

**NOTE:** AN EXTERNAL DISCONNECT SWITCH MUST BE INSTALLED AT THE TIME OF UNIT INSTALLATION, LOCATED WITHIN 6 FEET (APPROXIMATELY 2 METERS) OF THE UNIT.

## TABLE OF CONTENTS

|                                                              |    |
|--------------------------------------------------------------|----|
| 1 IMPORTANT SAFETY INSTRUCTIONS.....                         | 4  |
| 2 SHIPPING INSPECTION .....                                  | 4  |
| 2.1 PARTS .....                                              | 4  |
| 2.2 HANDLING.....                                            | 4  |
| 2.3 SHIPPING MATERIAL REMOVAL .....                          | 4  |
| 3 CODES & REGULATIONS.....                                   | 4  |
| 4 REPLACEMENT PARTS .....                                    | 5  |
| 5 PRE-INSTALLATION INSTRUCTIONS .....                        | 5  |
| 5.1 PREPARATION .....                                        | 5  |
| 5.2 SYSTEM MATCHES .....                                     | 5  |
| 5.3 CLEARANCES .....                                         | 5  |
| 5.4 INTERCONNECTING TUBING .....                             | 5  |
| 5.5 STANDING PRESSURE TEST (RECOMMENDED BEFORE VACCUM) ..... | 6  |
| 5.6 DEEP VACUUM METHOD .....                                 | 6  |
| 5.7 BRAZING .....                                            | 6  |
| 5.8 ACCESS.....                                              | 7  |
| 6 APPLICATION INFORMATION.....                               | 7  |
| 7 ELECTRIC HEAT .....                                        | 7  |
| 8 ELECTRICAL AND CONTROL WIRING .....                        | 8  |
| 8.1 BUILDING ELECTRICAL SERVICE INSPECTION .....             | 8  |
| 8.2 WIRE SIZING .....                                        | 8  |
| 8.3 MAXIMUM OVERCURRENT PROTECTION (MOP) .....               | 9  |
| 8.4 ELECTRICAL CONNECTIONS – SUPPLY VOLTAGE.....             | 9  |
| 8.5 FRESH AIR OPENINGS .....                                 | 9  |
| 8.6 WIRING .....                                             | 9  |
| 8.7 LOW VOLTAGE CONNECTIONS .....                            | 10 |
| 9 AIRFLOW TRIM.....                                          | 10 |
| 10 ACHIEVING LOW AIR LEAKAGE RATE .....                      | 10 |
| 11 MISCELLANEOUS START-UP CHECKLIST.....                     | 10 |
| 11.1 AUXILIARY ALARM SWITCH.....                             | 11 |
| 11.2 CIRCULATOR BLOWER.....                                  | 11 |
| 11.3 ACCESSORY CONTACTS .....                                | 12 |
| 11.4 LEAK DETECTION OUTPUT (REPLAY K4R).....                 | 13 |
| 12 TROUBLESHOOTING.....                                      | 13 |
| 12.1 ELECTROSTATIC DISCHARGE (ESD) PRECAUTIONS .....         | 13 |
| 12.2 DIAGNOSTIC CHART.....                                   | 14 |
| 12.3 FAULT RECALL .....                                      | 14 |
| 13 FULLY COMMUNICATING DAIKIN OR GOODMAN SYSTEM .....        | 14 |

|                                                                        |    |
|------------------------------------------------------------------------|----|
| 13.1 OVERVIEW.....                                                     | 14 |
| 13.2 AIRFLOW CONSIDERATION .....                                       | 14 |
| 13.3 THERMOSTAT WIRING.....                                            | 15 |
| 13.3.1 TWO-WIRE OUTDOOR AND FOUR-WIRE INDOOR WIRING .....              | 15 |
| 13.4 NETWORK TROUBLESHOOTING.....                                      | 15 |
| 13.5 SYSTEM TROUBLESHOOTING.....                                       | 15 |
| 13.6 REFRIGERATION DETECTION SYSTEM (RDS) .....                        | 16 |
| 13.6.1 INSTRUCTION TO REPLACE PCB.....                                 | 16 |
| 13.6.2 INSTRUCTION TO REPLACE A2L SENSOR .....                         | 17 |
| 13.6.2.1 SENSOR DISMOUNTING INSTRUCTIONS.....                          | 17 |
| 13.6.2.2 SENSOR MOUNTING INSTRUCTIONS .....                            | 17 |
| 13.7 MITIGATION AIRFLOW .....                                          | 17 |
| 13.7.1 ALTITUDE ADJUSTMENT FACTOR TO CALCULATE MINIMUM ROOM AREA ..... | 18 |
| 13.8 REFRIGERANT LEAK TEST.....                                        | 18 |
| 13.9 DETECTION OF REFRIGERANT.....                                     | 19 |
| 13.10 REMOVAL AND EVACUATION.....                                      | 19 |
| 13.11 REFRIGERANT RECOVERY .....                                       | 19 |
| 14 TROUBLESHOOTING.....                                                | 20 |
| 15 NETWORK TROUBLESHOOTING.....                                        | 22 |
| 16 DIAGNOSTIC CODES .....                                              | 23 |
| 17 SETTING THE MODE DISPLAY .....                                      | 24 |
| 18 WIRING DIAGRAM.....                                                 | 26 |
| 19 START-UP CHECKLIST .....                                            | 29 |



### WARNING

DO NOT USE MEANS TO ACCELERATE THE DEFROSTING PROCESS OR TO CLEAN, OTHER THAN THOSE RECOMMENDED BY THE MANUFACTURER. THE APPLIANCE SHALL BE STORED IN A ROOM WITHOUT CONTINUOUSLY OPERATING IGNITION SOURCES (FOR EXAMPLE: OPEN FLAMES, AN OPERATING GAS APPLIANCE OR AN OPERATING ELECTRIC HEATER). DO NOT PIERCE OR BURN. BE AWARE THAT REFRIGERANTS MAY NOT CONTAIN AN ODOR.



### WARNING

A REFRIGERANT LEAK DETECTION SYSTEM IS INSTALLED. THE UNIT MUST BE POWERED ON AT ALL TIMES EXCEPT FOR SERVICE.



### WARNING

**HIGH VOLTAGE**  
DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



### WARNING

TO PREVENT THE RISK OF PROPERTY DAMAGE, PERSONAL INJURY, OR DEATH, DO NOT STORE COMBUSTIBLE MATERIALS OR USE GASOLINE OR OTHER FLAMMABLE LIQUIDS OR VAPORS IN THE VICINITY OF THIS APPLIANCE.



### CAUTION

HAVE YOUR CONTRACTOR IDENTIFY ALL THE VARIOUS CUT OFF SWITCHES AND DEVICES THAT SERVICE THIS UNIT. KNOW WHERE THE SWITCH IS THAT WILL CUT OFF ENERGY TO THE HEATING SYSTEM IN THE EVENT OF OVERHEATING.



## WARNING

PARTIAL UNITS SHALL ONLY BE CONNECTED TO AN APPLIANCE SUITABLE FOR THE SAME REFRIGERANT. THIS UNIT IS A PARTIAL UNIT AIR CONDITIONER, COMPLYING WITH PARTIAL UNIT REQUIREMENTS OF THIS INTERNATIONAL STANDARD, AND MUST ONLY BE CONNECTED TO OTHER UNITS THAT HAVE BEEN CONFIRMED AS COMPLYING TO CORRESPONDING PARTIAL UNIT REQUIREMENTS OF THIS INTERNATIONAL STANDARD.



## WARNING

THIS UNIT SHOULD NOT BE CONNECTED TO, OR USED IN CONJUNCTION WITH, ANY DEVICES THAT ARE NOT CERTIFIED FOR USE WITH THIS UNIT OR HAVE NOT BEEN TESTED AND APPROVED BY THE MANUFACTURER. SERIOUS PROPERTY DAMAGE OR PERSONAL INJURY, REDUCED UNIT PERFORMANCE AND/OR HAZARDOUS CONDITIONS MAY RESULT FROM THE USE OF DEVICES THAT HAVE NOT BEEN APPROVED OR CERTIFIED BY THE MANUFACTURER.

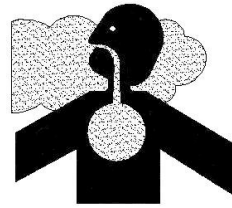


## WARNING

APPLIANCE SHOULD NOT BE ACCESSIBLE TO THE GENERAL PUBLIC. CHECK SERIAL PLATE FOR MAXIMUM OPERATING CONDITIONS.



## DANGER PELIGRO



### CARBON MONOXIDE POISONING HAZARD

**Special Warning for Installation of Furnace or Air Handling Units in Enclosed Areas such as Garages, Utility Rooms or Parking Areas**

Carbon monoxide producing devices (such as automobiles, space heater, gas water heater, etc.) should not be operated in enclosed areas such as unventilated garages, utility rooms or parking areas because of the danger of carbon monoxide (CO) poisoning resulting from the exhaust emissions. If a furnace or air handler is installed in an enclosed area such as a garage, utility room or parking area and a carbon monoxide producing device is operated therein, there must be adequate, direct outside ventilation.

This ventilation is necessary to avoid the danger of CO poisoning which can occur if a carbon monoxide producing device continues to operate in the enclosed area. Carbon monoxide emissions can be (re)circulated throughout the structure if the furnace or air handler is operating in any mode.

CO can cause serious illness including permanent brain damage or death.

B10259-216

### RIESGO DE INTOXICACIÓN POR MONÓXIDO DE CARBONO

Advertencia especial para la instalación de calentadores ó manejadoras de aire en áreas cerradas como estacionamientos ó cuartos de servicio.

Los equipos ó aparatos que producen monóxido de carbono (tal como automóvil, calentador de gas, calentador de agua por medio de gas, etc) no deben ser operados en áreas cerradas debido al riesgo de envenenamiento por monóxido de carbono (CO) que resulta de las emisiones de gases de combustión. Si el equipo ó aparato se opera en dichas áreas, debe existir una adecuada ventilación directa al exterior.

Esta ventilación es necesaria para evitar el peligro de envenenamiento por CO, que puede ocurrir si un dispositivo que produce monóxido de carbono sigue operando en el lugar cerrado.

Las emisiones de monóxido de carbono pueden circular a través del aparato cuando se opera en cualquier modo.

El monóxido de carbono puede causar enfermedades severas como daño cerebral permanente ó muerte.

B10259-216

### RISQUE D'EMPOISONNEMENT AU MONOXYDE DE CARBONE

Avertissement special au sujet de l'installation d'appareils de chauffage ou de traitement d'air dans des endroits clos, tels les garages, les locaux d'entretien et les stationnementss.

Evitez de mettre en marche les appareils produisant du monoxyde de carbone (tels que les automobile, les appareils de chauffage autonome, etc.) dans des endroits non ventilés tels que les d'empoisonnement au monoxyde de carbone. Si vous devez faire fonctionner ces appareils dans un endroit clos, assurez-vous qu'il y ait une ventilation directe provenant de l'exterieur.

Cette ventilation est nécessaire pour éviter le danger d'intoxication au CO pouvant survenir si un appareil produisant du monoxyde de carbone continue de fonctionner au sein de la zone confinée.

Les émissions de monoxyde de carbone peuvent etre recircules dans les endroits clos, si l'appareil de chauffage ou de traitement d'air sont en marche.

Le monoxyde de carbone peut causer des maladies graves telles que des dommages permanents au cerveau et meme la mort.

B10259-216



## WARNING

FAILURE TO PROPERLY RECONNECT SENSOR WIRES MAY RESULT IN ERROR CODES AND THE UNIT NOT OPERATING.



## WARNING

AUXILIARY DEVICES WHICH MAY BE A POTENTIAL IGNITION SOURCE SHALL NOT BE INSTALLED IN THE DUCT WORK. EXAMPLES OF SUCH POTENTIAL IGNITION SOURCES ARE HOT SURFACES WITH A TEMPERATURE EXCEEDING 650 °C AND ELECTRIC SWITCHING DEVICES. FOR EXAMPLE ELECTRONIC AIR FILTERS ARE A POTENTIAL IGNITION SOURCE.



## WARNING

ONLY AUXILIARY DEVICES APPROVED BY THE APPLIANCE MANUFACTURER OR DECLARED SUITABLE WITH THE REFRIGERANT SHALL BE INSTALLED IN CONNECTING DUCTWORK. IF AN EXISTING DEVICE IS NOT APPROVED BY THE APPLIANCE MANUFACTURER FOR USE WITH THE REFRIGERANT, THE DEVICE MUST BE REMOVED OR DISCONNECTED.



## WARNING

IF ANY HOT WORK IS TO BE CONDUCTED ON THE REFRIGERATING EQUIPMENT OR ANY ASSOCIATED PARTS, APPROPRIATE FIRE EXTINGUISHING EQUIPMENT SHALL BE AVAILABLE ON HAND. HAVE A DRY POWDER OR CO<sub>2</sub> FIRE EXTINGUISHER ADJACENT TO THE HOT WORK AREA.

ENSURE THAT THE AREA IS IN THE OPEN OR THAT IT IS ADEQUATELY VENTILATED BEFORE BREAKING INTO THE SYSTEM OR CONDUCTING ANY HOT WORK. A DEGREE OF VENTILATION SHALL CONTINUE DURING THE PERIOD THAT THE WORK IS CARRIED OUT. THE VENTILATION SHOULD SAFELY DISPERSE ANY RELEASED REFRIGERANT AND PREFERABLY EXPEL IT EXTERNALLY INTO THE ATMOSPHERE.



## WARNING

ALL ACCESSORIES THAT MAY BECOME A POTENTIAL IGNITION SOURCE IF INSTALLED, SUCH AS ELECTRONIC AIR CLEANERS, MUST ONLY BE POWERED THROUGH OUR ACCESSORY CONTROL BOARD KIT. IF AN ELECTRONIC AIR CLEANER IS ALREADY INSTALLED IN THE DUCT WORK AND NOT CONNECTED TO THE ACCESSORY CONTROL BOARD, IT WILL HAVE TO BE DISABLED OR REMOVED. ENSURE THAT ANY ADDITIONAL WIRING FROM THE INDOOR UNIT TO THE ACCESSORY CONTROL BOARD IS ROUTED AND PROTECTED FROM DAMAGE AND WEAR, AVOIDING THE FLUE PIPE AND ANY JOINTS THAT MAY NEED BRAZED OR DISCONNECTED FOR SERVICE. REFER TO THE PRODUCT SPECIFICATION SHEET FOR THE ACCESSORY CONTROL BOARD KIT PART NUMBER.

**NOTE: READ THE ENTIRE INSTRUCTION MANUAL BEFORE STARTING THE INSTALLATION.**

## 1 IMPORTANT SAFETY INSTRUCTIONS

Symbols and labels are used throughout this manual to indicate immediate or potential safety hazards. It is the owner's and installer's responsibility to read and comply with all safety information and instructions accompanying these symbols. Failure to heed safety information increases the risk of personal injury, property damage, and/or product damage.

**NOTICE: THIS PRODUCT CONTAINS ELECTRIC COMPONENTS WHICH REQUIRE A DEFINITE GROUND. PROVISIONS ARE MADE FOR CONNECTION OF THE GROUND. A DEDICATED GROUND FROM THE MAIN POWER SUPPLY OR AN EARTH GROUND MUST BE PROVIDED.**

## 2 SHIPPING INSPECTION

Upon receiving the product, inspect it for damage from shipment. Shipping damage, and subsequent investigation is the responsibility of the carrier. Verify the model number, specifications, electrical characteristics, and accessories are correct prior to installation. The distributor or manufacturer will not accept claims from dealers for transportation damage or installation of incorrectly shipped units.

### 2.1 PARTS

Inspect the unit to verify all required components are present and intact. Report any missing components immediately to the manufacturer or to the distributor. Use only factory authorized replacement parts (see Section 4). Make sure to include the full product model number and serial number when reporting and/or obtaining service parts.

### 2.2 HANDLING

Use caution when transporting / carrying the unit. Do not move unit using shipping straps. Do not carry unit with hooks or sharp objects. The preferred method of carrying the unit after arrival at the job site is to carry via a two-wheel hand truck from the back or sides or via hand by carrying at the cabinet corners. If carrying by hand, carry at the cabinet corners with two people. Avoid holding the unit by the piping.

### 2.3 SHIPPING MATERIAL REMOVAL

**IMPORTANT: ALL SHIPPING MATERIAL USED TO PROTECT THE EQUIPMENT, AND THE EQUIPMENT'S COMPONENTS, DURING TRANSIT SHOULD BE REMOVED BEFORE FINAL INSTALLATION.**

## 3 CODES & REGULATIONS

This product is designed and manufactured to comply with national codes such as ASHRAE 15, ASHRAE 15.2, IAPMO Uniform Mechanical Code, ICC International Mechanical Code, or CSA B52. Installation in accordance with such codes and/or prevailing local codes/regulations is the responsibility of the installer. The manufacturer assumes no responsibility for equipment installed in violation of any codes or regulations.

The standard test ambient operating conditions for cooling and heating mode are 80°F and 70°F, respectively. **The United States Environmental Protection Agency (EPA) has issued various regulations regarding the introduction and disposal of refrigerants. Failure to follow these regulations may harm the environment and can lead to the imposition of substantial fines.** These regulations may vary by jurisdiction. A certified technician must perform the installation and service of this product. Should you have any questions please contact the local office of the EPA and/or refer to the EPA's website [www.epa.gov](http://www.epa.gov).

## 4 REPLACEMENT PARTS

When reporting shortages or damages, or ordering repair parts, give the complete product model and serial numbers as on the unit's Serial and Rating plate. Replacement parts for this product are available through your contractor or local distributor. Your nearest distributor can be located online at <https://partner.goodmanmfg.com/> or by contacting:

HOMEOWNER SUPPORT  
DAIKIN COMFORT TECHNOLOGIES  
MANUFACTURING, INC.  
19001 KERMIER ROAD  
WALLER, TEXAS 77484  
(855) 770-5678

### Installer's Note:

Replacement air filters can be ordered directly from the supplier:

United Air Filter  
Charlotte, NC  
Phone: **704-334-5311**

AWVE 12,18,24 - Size: 14" x 18" x 1"  
AWVE 30-36 - Size: 18" x 20" x 1"

## 5 PRE-INSTALLATION INSTRUCTIONS

### 5.1 PREPARATION

Keep this document with the unit. Carefully read all instructions for the installation prior to installing the product. Make sure each step or procedure is understood and any special considerations are taken into account before starting installation. Assemble all tools, hardware, and supplies needed to complete the installation. Some items may need to be purchased locally. Make sure everything needed to install the product is on hand before starting.

**NOTE: THESE AIR HANDLERS ARE DESIGNED FOR INDOOR INSTALLATION ONLY AT A MAX ALTITUDE OF 10,500 FEET ABOVE SEA LEVEL OR A MIN ALTITUDE OF -184 FEET BELOW SEA LEVEL.**

**NOTE: IF THE UNIT IS LOCATED IN AN UNCONDITIONED AREA WITH HIGH AMBIENT TEMPERATURE AND/OR HIGH HUMIDITY, THE AIR HANDLER MAY BE SUBJECT TO NUISANCE SWEATING OF THE CASING. ON THESE INSTALLATIONS, A WRAP OF 2" FIBERGLASS INSULATION WITH A VAPOR BARRIER IS RECOMMENDED. DO NOT COVER WARNING LABELS OR SERIAL PLATE. A SECONDARY DRAIN PAN BELOW THE UNIT IS ALSO RECOMMENDED TO PROTECT THE INSTALLATION SITE.**

### 5.2 SYSTEM MATCHES

The entire system (combination of indoor and outdoor sections) must be manufacture approved and Air-conditioning, Heating, and Refrigeration Institute (AHRI) listed.

**NOTE: INSTALLATION OF UNMATCHED SYSTEMS IS NOT PERMITTED. DAMAGE OR REPAIRS DUE TO INSTALLATION OF UNMATCHED SYSTEMS IS NOT COVERED UNDER THE WARRANTY.**

### 5.3 CLEARANCES

|                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>WARNING</b>                                                                                                                                                                              |
| WHEN INSTALLED IN A ROOM WITH AN AREA LESS THAN THAT OUTLINED IN TABLE 13, THAT ROOM SHALL BE WITHOUT CONTINUOUSLY OPERATING OPEN FLAMES (FOR EXAMPLE AN OPERATING GAS APPLIANCE) OR OTHER POTENTIAL IGNITION SOURCES (FOR EXAMPLE AN OPERATING ELECTRIC HEATER, HOT SURFACES). |

The unit clearance from a combustible surface may be 0". However, service clearance must take precedence. A minimum of 24" in front of the unit for service clearance is required. Additional clearance on one side or top will be required for electrical wiring connections. Consult all appropriate regulatory codes prior to determining final clearances. When installing this unit in an area that may become wet (such as crawl spaces), elevate the unit with a sturdy, non-porous material. In installations that may lead to physical damage (i.e. a garage) it is advised to install a protective barrier to prevent such damage. Always install units such that a positive slope in condensate line (1/4" per foot) is allowed. Further, any joints made in the installation between parts of the refrigerating system must be accessible for maintenance purposes.

### 5.4 INTERCONNECTING TUBING

|                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>WARNING</b>                                                                                          |
| THIS PRODUCT IS FACTORY-SHIPED WITH DRY NITROGEN MIXTURE GAS UNDER PRESSURE. USE APPROPRIATE SERVICE TOOLS AND FOLLOW THESE INSTRUCTIONS TO PREVENT INJURY.                                   |
|  <b>WARNING</b>                                                                                          |
| A QUENCHING CLOTH IS STRONGLY RECOMMENDED TO PREVENT SCORCHING OR MARRING OF THE EQUIPMENT FINISH WHEN BRAZING CLOSE TO THE PAINTED SURFACES. USE BRAZING ALLOY OF 5% MINIMUM SILVER CONTENT. |





## WARNING

WORK SHALL BE UNDERTAKEN UNDER A CONTROLLED PROCEDURE TO MINIMIZE RISK OF FLAMMABLE GAS OR VAPOR BEING PRESENT WHILE THE WORK IS BEING PERFORMED.



## WARNING

ONLY BRAZING TECHNIQUES AND APPROVED MECHANICAL JOINTS SHOULD BE USED TO CONNECT REFRIGERANT TUBING CONNECTIONS. NON-APPROVED MECHANICAL CONNECTORS AND OTHER METHODS ARE NOT PERMITTED IN THIS SYSTEM CONTAINING R-32 REFRIGERANT. APPROVED MECHANICAL JOINTS WILL BE DETAILED IN THE PRODUCTS SPECIFICATIONS SHEET.

Give special consideration to minimize the length of refrigerant tubing when installing air handlers. Refer to Remote Cooling/ Heat Pump Technical Publication TP-110\* Long Line Set Application R-32 for tubing guidelines. If possible, allow adequate length of tubing such that the coil may be removed (for inspection or cleaning services) from the cabinet without disconnecting the tubing.

Indoor equipment and pipes must be securely mounted and guarded such that accidental rupture of equipment cannot occur from such events like moving furniture or reconstruction activities. Field-made indoor refrigerant joints shall be leak tested.

### 5.5 STANDING PRESSURE TEST (RECOMMENDED BEFORE VACUUM)

Field pipework must be pressure tested with an inert gas and then vacuum tested prior to refrigerant charging according to the following requirements:

Pressurize the system to the maximum allowable pressure listed in the serial plate using dry nitrogen or dry helium. Allow the pressure to stabilize and hold for at least 15 minutes. The system is considered leak-free if the pressure does not drop below the selected maximum allowable pressure. If, after 15 minutes, the pressure drops, it implies a leak in the system. Proceed with identifying and sealing the leak and repeating the Standing Pressure Test. Proceed to system evacuation using the Deep Vacuum Method. Leak test the system using dry nitrogen or dry helium and soapy water to identify leaks. **NO REFRIGERANT SHALL BE USED FOR PRESSURE TESTING TO DETECT LEAKS.** Proceed to system evacuation using the Deep Vacuum Method.

Field-made refrigerant joints indoors shall be tightness tested. The test method shall have a sensitivity of 5 grams (0.176 oz) per year of refrigerant or better under a pressure of at least 0.25 times the maximum allowable pressure. No leak shall be detected.

### 5.6 DEEP VACUUM METHOD

The Deep Vacuum Method requires a vacuum pump rated for 500 microns or less. This method effectively and efficiently ensures the system is free of non-condensable air and moisture. The Triple Evacuation Method is detailed in the Service Manual for this product model as an alternative. To expedite the evacuation procedure, it is recommended that the Schrader Cores be removed from the service valves using a core-removal tool.

1. Connect the vacuum pump, micron gauge, and vacuum-rated hoses to both service valves. Evacuation must use both service valves to eliminate system mechanical seals.
2. Evacuate the system to less than 500 microns.
3. Isolate the pump from the system and hold the vacuum for 10 minutes (minimum). Typically, pressure will rise slowly during this period. If the pressure rises to less than 1000 microns and remains steady, the system is considered leak-free; proceed to system charging and startup.
4. If pressure rises above 1000 microns but holds steady below 2000 microns, non-condensable air or moisture may remain, or a small leak may be present. Return to step 2: If the same result is achieved, check for leaks and repair. Repeat the evacuation procedure.
5. If pressure rises above 2000 microns, a leak is present. Check for leaks and repair them. Then, repeat the evacuation procedure.

### 5.7 BRAZING

Braze joints should be made only with the connections provided external to the cabinet. Do not alter the cabinet nor braze inside the cabinet. To avoid overheating after brazing, quench all brazed joints with water or a wet rag.

This system must be charged with R-32 refrigerant in order to operate safely.

When breaking into the refrigerant circuit to make repairs - or for any other purpose - conventional procedures shall be used. However, the flammability classification for any refrigerant must be considered. The following procedure shall be adhered to:

1. Safely remove refrigerant following local and national regulations.
2. Evacuate.
3. Purge the circuit with inert gas.
4. Evacuate.
5. Continuously flush or purge with inert gas when using flame to open circuit.
6. Open the circuit.

For appliances containing flammable refrigerants, refrigerants purging shall be achieved by breaking the vacuum in the system with oxygen-free nitrogen and continuing to fill until the working pressure is achieved, then venting to atmosphere, and finally pulling down to a vacuum (optional for R-32). This process shall be repeated until no refrigerant is within the system (optional for R-32). When the final oxygen-free nitrogen charge is used, the system shall be vented down to atmospheric pressure to enable work to take place.

The outlet for the vacuum pump shall not be close to any potential ignition sources, and ventilation shall be available.



## CAUTION

APPLYING TOO MUCH HEAT TO ANY TUBE CAN MELT THE TUBE. TORCH HEAT REQUIRED TO BRAZE TUBES OF VARIOUS SIZES MUST BE PROPORTIONAL TO THE SIZE OF THE TUBE. SERVICE PERSONNEL MUST USE THE APPROPRIATE HEAT LEVEL FOR THE SIZE OF THE TUBE BEING BRAZED.

### 5.8 ACCESS

This unit should be installed in a manner so that it is not accessible to the public.

## 6 APPLICATION INFORMATION

The unit is designed to be installed in conditioned space, either recessed into a wall or hanging in a vertical “upflow” position. If units are recessed in a wall, use the holes along the inside of the front flange to attach units to the framing studs. The vertical air handler comes equipped with an offset hanging bracket attached to the rear of the cabinet for hanging applications.

**IMPORTANT: UNIT MUST BE INSTALLED LEVEL OR SLIGHTLY TILTED BACK TO ENSURE PROPER DRAINAGE OF CONDENSATE.**

The air handler also has a bottom and front return. **Large chassis installations should be installed as front return only.** If the small chassis air handler is to be installed in a bottom return application, discard the drain access panel in the bottom of the unit. If the air handler is to be installed in a front return application, remove and discard the front access panel with insulation. The unit is shipped with a filter rack and filter. Remember to inspect, clean and/or replace the filter monthly.

Units are equipped with both a bottom primary and secondary drain. Both drains must be trapped, unless emergency condensate switch is install in the secondary drain. Failure to install a trap could result in condensation overflowing the drain pan resulting in substantial water damage to the nearby area.

The connectors required are 3/4” NPT male, either PVC or metal pipe, CPVC piping is not approved, and should be hand tightened to a torque of no more than 37 in-lbs. to prevent damage to the drain pan connection.

To prevent potential sweating and dripping on to finished space, it may be necessary to insulate the condensate drain line located inside the building. Use Armaflex® or similar material.

**NOTE: IF YOU INTEND TO INSTALL THIS UNIT WITH A “WAD” DOOR IT MUST BE MOUNTED 1/4” BEHIND FRONT EDGE OF STUD.**

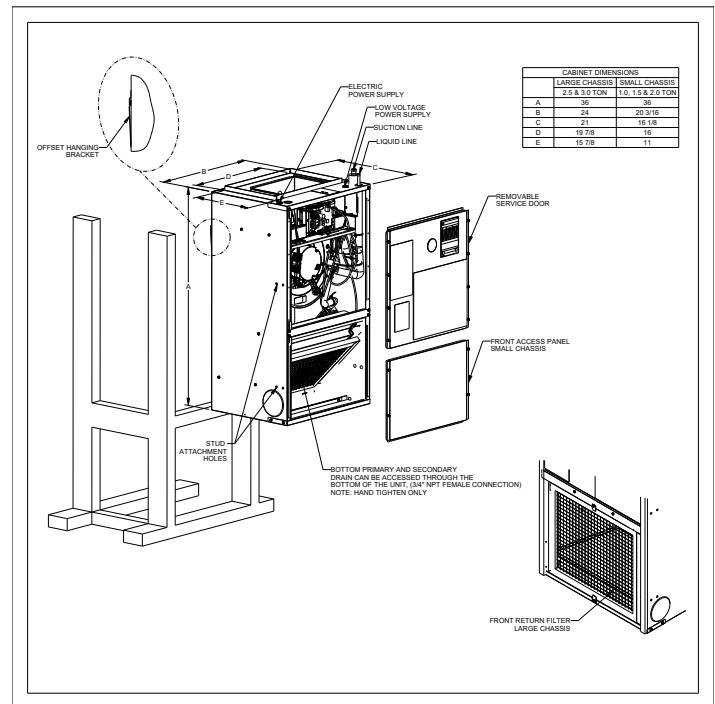


FIGURE 1

## 7 ELECTRIC HEAT

Refer to the installation manual provided with the electric heat kit for the correct installation procedure. All electric heat must be field installed. If installing this option, the **ONLY** heat kits that are permitted to be used are the Daikin produced HKTW series. Refer to the air handler unit's Serial and Rating plate to determine the heat kits compatible with a given air handler. No other accessory heat kit besides the HKTW series may be installed in these air handlers. Refer to Table 4 for Absolute minimum allowable airflow for Airhandler and heat kit combination.

The heating mode temperature rise is dependent upon the system airflow, the supply voltage, and the heat kit size (kW) selected. Use data provided in Tables 1,2, and 3 to determine the temperature rise (°F).

**NOTE:** For emergency heat, set the dipswitch on PCB. For heating mode, use the thermostat user menu. For installations not indicated above the following formula is to be used:

$$TR = (kW \times 3412) \times (\text{Voltage Correction}) / (1.08 \times CFM)$$

Where:

- TR = Temperature Rise
- kW = Heater Kit Actual kW
- 3412 = Btu per kW
- VC\* = 1.0 (240 Supply Volts)
- = .92 (230 Supply Volts)

= .84 (220 Supply Volts)  
 = .77 (210 Supply Volts)  
 = .75 (208 Supply Volts)  
 1.08 = Constant  
 CFM = Measured Airflow  
 \*VC (Voltage Correction)

| CFM  | HEATER (kW)<br>3 |
|------|------------------|
| 600  | 16               |
| 800  | 12               |
| 1000 | 9                |
| 1200 | 8                |

240/1/60 SUPPLY VOLTAGE - TEMP. RISE °F  
TABLE 1

| CFM  | HEATER (kW)<br>3 |
|------|------------------|
| 600  | 15               |
| 800  | 11               |
| 1000 | 9                |
| 1200 | 7                |

230/1/60 SUPPLY VOLTAGE - TEMP. RISE °F  
TABLE 2

| CFM  | HEATER (kW)<br>3 |
|------|------------------|
| 600  | 12               |
| 800  | 9                |
| 1000 | 7                |
| 1200 | 6                |

208/1/60 SUPPLY VOLTAGE - TEMP. RISE °F  
TABLE 3

| MODEL        | HEATER CFM<br>3 (kW) |
|--------------|----------------------|
| AWVE12SU1300 | NOT RATED            |
| AWVE18SU1300 | NOT RATED            |
| AWVE24SU1300 | NOT RATED            |
| AWVE30LU1300 | 596                  |
| AWVE36LU1300 | 596                  |

TABLE 4

**NOTE: AIRFLOWS SHOWN ARE TO BE CONSIDERED THE ABSOLUTE MINIMUM ALLOWABLE FOR THE AIR HANDLER AND HEAT KIT COMBINATION. THE MINIMUM AIRFLOW DOES NOT REPRESENT THE RECOMMENDED AIRFLOW BY THE MANUFACTURER.**

## 8 ELECTRICAL AND CONTROL WIRING

**IMPORTANT: THE APPLIANCE SHALL BE INSTALLED IN ACCORDANCE WITH NATIONAL WIRING REGULATIONS. ALL ROUTING OF ELECTRICAL WIRING MUST BE MADE THROUGH PROVIDED ELECTRICAL KNOCKOUTS. DO NOT CUT, PUNCTURE OR ALTER THE CABINET FOR ELECTRICAL WIRING.**



### WARNING

#### HIGH VOLTAGE

TO AVOID THE RISK OF FIRE OR EQUIPMENT DAMAGE, USE ONLY COPPER CONDUCTORS. DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. THE ELECTRICAL POWER TO THIS UNIT MUST BE IN THE OFF POSITION AND ALL POWER SUPPLIES DISCONNECTED. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.



### WARNING

THE UNIT MUST HAVE AN UNINTERRUPTED, UNBROKEN ELECTRICAL GROUND TO MINIMIZE THE POSSIBILITY OF PERSONAL INJURY IF AN ELECTRICAL FAULT SHOULD OCCUR. THE ELECTRICAL GROUND CIRCUIT MAY CONSIST OF AN APPROPRIATELY SIZED ELECTRICAL WIRE CONNECTING THE GROUND LUG IN THE UNIT AND CONTROL BOX WIRE TO THE BUILDING'S ELECTRICAL SERVICE PANEL. OTHER METHODS OF GROUNDING ARE PERMITTED IF PERFORMED IN ACCORDANCE WITH THE "NATIONAL ELECTRIC CODE" (NEC)/"AMERICAN NATIONAL STANDARDS INSTITUTE" (ANSI)/"NATIONAL FIRE PROTECTION ASSOCIATION" (NFPA) 70 AND LOCAL/STATE CODES. IN CANADA, ELECTRICAL GROUNDING IS TO BE IN ACCORDANCE WITH THE CANADIAN ELECTRIC CODE CSA C22.1. FAILURE TO OBSERVE THIS WARNING CAN RESULT IN ELECTRICAL SHOCK THAT CAN CAUSE PERSONAL INJURY OR DEATH.

### 8.1 BUILDING ELECTRICAL SERVICE INSPECTION

This unit is designed for single-phase electrical supply.

**NOTE: DO NOT OPERATE ON A THREE-PHASE POWER SUPPLY.**

Measure the power supply to the unit. The supply voltage must be in agreement with the unit name plate power requirements and within the range shown in Table 5.

| Nominal Input | Minimum Voltage | Maximum Voltage |
|---------------|-----------------|-----------------|
| 208/230       | 187             | 253             |

ELECTRICAL VOLTAGE  
TABLE 5

### 8.2 WIRE SIZING

Wire size is important to the operation of your equipment. Use the following checklist when selecting the appropriate wire size for the unit.

- Wire used must carry the Minimum circuit Ampacity (MCA) listed on the unit's Serial and Rating Plate.
- Refer to the NEC (USA) or CSA (Canada) for wire sizing. The unit MCA for the air handler and the optional electric heat kit can be found on the unit Serial and Rating Plate.
- Wire must be sized to allow no more than a 2% voltage drop from the building breaker/fuse panel to the unit.
- Wires with different insulation temperature rating have varying ampacities - be sure to check the temperature rating used.



Refer to the latest edition of the National Electric Code or in Canada the Canadian Electric Code when determining the correct wire size.

### 8.3 MAXIMUM OVERCURRENT PROTECTION (MOP)

Every installation must include an NEC (USA) or CEC (Canada) approved overcurrent protection device. Also, check with local or state codes for any special regional requirements.

Protection can be in the form of fusing or HACR style circuit breakers. The Serial and Rating Plate can be used as a guide for selecting the MAXIMUM overcurrent device.

**NOTE: FUSES OR CIRCUIT BREAKERS ARE TO BE SIZED LARGER THAN THE EQUIPMENT MCA BUT NOT TO EXCEED THE MOP.**

**IMPORTANT NOTE: TORCH HEAT REQUIRED TO BRAZE TUBES OF VARIOUS SIZES IS PROPORTIONAL TO THE SIZE OF THE TUBE. TUBES OF SMALLER SIZE REQUIRE LESS HEAT TO BRING THE TUBE TO BRAZING TEMPERATURE BEFORE ADDING BRAZING ALLOY. APPLYING TOO MUCH HEAT TO ANY TUBE CAN MELT THE TUBE. SERVICE PERSONNEL MUST USE THE APPROPRIATE HEAT LEVEL FOR THE SIZE OF THE TUBE BEING BRAZED.**

**NOTE: THE USE OF A HEAT SHIELD WHEN BRAZING IS RECOMMENDED TO AVOID BURNING THE SERIAL PLATE OR THE FINISH ON THE UNIT.**

### 8.4 ELECTRICAL CONNECTIONS – SUPPLY VOLTAGE



#### CAUTION

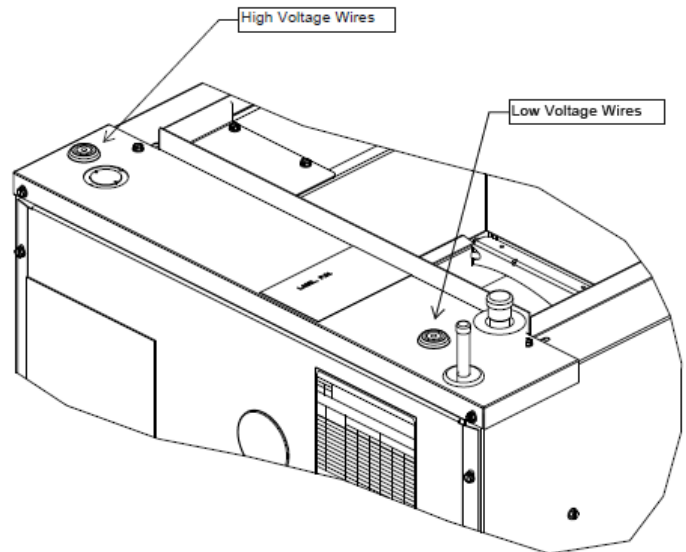
##### FIRE HAZARD!

TO AVOID THE RISK OF PROPERTY DAMAGE, PERSONAL INJURY OR FIRE, USE ONLY COPPER CONDUCTORS.

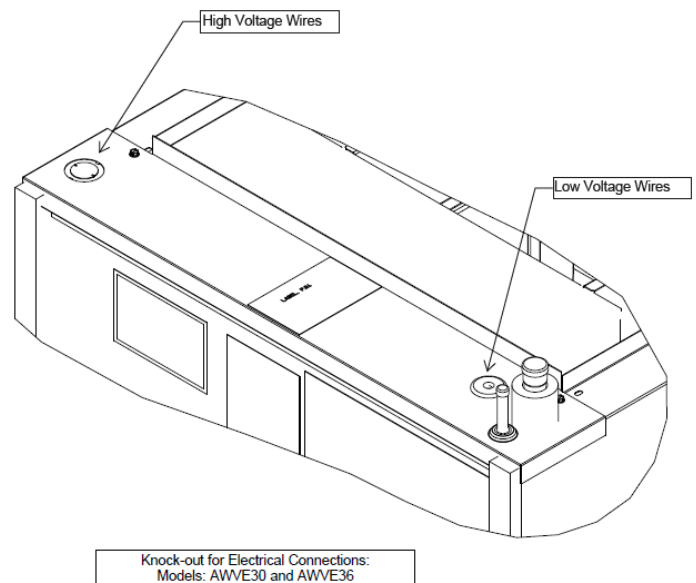
**IMPORTANT NOTE: USE COPPER CONDUCTORS ONLY FROM DISCONNECT OR ELECTRICAL PANEL TO THE AIR HANDLER.**

A knockout is provided on the air handler top panel. An NEC or CEC approved strain relief is to be used at this entry point. The wire is to be sized in accordance with the “Electrical Wire and MOP” section of this manual. Some areas require the supply wire to be enclosed in conduit. Consult your local codes.

**NOTE: AS SHOW IN FIGURES 2 AND 3 BELOW, HIGH VOLTAGE WIRES MUST BE SEPARATED DURING INSTALLATION.**



MODELS: AWVE12, 18, 24  
KNOCK-OUT FOR ELECTRICAL CONNECTIONS  
FIGURE 2



MODELS: AWVE30, 36  
KNOCK-OUT FOR ELECTRICAL CONNECTIONS  
FIGURE 3

### 8.5 FRESH AIR OPENINGS

All AWVE models include 4” fresh air knockouts on both sides of the cabinet. To utilize, simply remove the appropriate knockout and connect fresh air inlet venting directly to cabinet. Ensure the unit mounting system does not block the fresh air inlet holes.

### 8.6 WIRING

Check that cabling will not be subject to wear, corrosion, excessive pressure, vibration, sharp edges or any other adverse environmental effects. The check shall also take into account the effects of aging or continual vibration from sources such as compressors or fans.

**NOTE: THE INSTALLING CONTRACTOR MUST TAKE INTO ACCOUNT ANY LOST CAPACITY DUE TO THE FRESH AIR OPENINGS WHEN SIZING THE EQUIPMENT.**

## 8.7 LOW VOLTAGE CONNECTIONS

Several combinations of low voltage schemes are possible, depending on the presence of a heat kit and whether the heat kit is single-stage or multi-stage, whether the outdoor section is an air conditioner or heat pump, and whether the outdoor section is single-stage or two stage. The 24V-control voltage connects the air handler to the room thermostat and condenser. Low voltage wiring must be copper conductors. A minimum of 18AWG must be used for installations. Low voltage wiring must be connected through the top of the cabinet or either side.

## 9 AIRFLOW TRIM

Indoor airflow can be trimmed up/down through the outdoor unit user menu. For more detailed information, please refer to the outdoor unit installation manual.

## 10 ACHIEVING LOW AIR LEAKAGE RATE

Ensure all the gaskets remain intact on surfaces as shipped with the unit. When these requirements are satisfied, the unit achieves the following airflow leakage when tested in accordance with ASHRAE Standard 193:

- Cabinet air leakage less than 2% at 1.0 inch H<sub>2</sub>O
- Cabinet air leakage less than 1.4% at 0.5 inch H<sub>2</sub>O

**IMPORTANT: AFTER INSTALLING THE HEATER KIT, IT IS VERY IMPORTANT TO SEAL THE GAP BETWEEN THE UNIT ACCESS PANEL AND THE UNIT. PUTTY PASTE OR GASKET CAN BE USED TO SEAL THE GAP SO THAT AIR LEAKAGE CAN BE MINIMIZED.**

## 11 MISCELLANEOUS START-UP CHECKLIST

- Prior to start-up, ensure that all electrical wires are properly sized and all connections are properly tightened.
- All panels must be in place and secured. For Air Tight application, gasket must be positioned at prescribed locations to achieve 2% leakage.
- Tubing must be leak free.
- Condensate line must be trapped and pitched to allow for drainage.
- Auxiliary drain is installed when necessary and pitched to allow for drainage.
- Low voltage wiring is properly connected.
- Unit is protected from vehicular or other physical damage.
- Return air is not obtained from, nor are there any return air duct joints that are unsealed in, areas where there may be objectionable odors, flammable vapors or products of combustion such as carbon monoxide (CO), which may cause serious personal injury or death.

| RATED INDOOR AIRFLOW (SCFM) |      |      |      |      |      |      |      |      |
|-----------------------------|------|------|------|------|------|------|------|------|
| STATIC PRESSURE (IN. WC)    |      |      |      |      |      |      |      |      |
| ID UNIT                     | 0.1  | 0.2  | 0.3  | 0.4  | 0.5  | 0.6  | 0.7  | 0.8  |
| AWVE12SU1300                | 450  | 450  | 450  | 450  | 450  | 450  | 450  | 450  |
| AWVE18SU1300                | 600  | 600  | 600  | 600  | 600  | 600  | 600  | 600  |
| AWVE24SU1300                | 830  | 830  | 830  | 830  | 830  | 830  | 830  | 830  |
| AWVE30LU1300                | 1000 | 1000 | 1000 | 1000 | 1000 | 1000 | 1000 | 1000 |
| AWVE36LU1300                | 1150 | 1150 | 1150 | 1150 | 1150 | 1150 | 1150 | 1150 |

TABLE 6

### NOTE:

1. AIRFLOW DATA INDICATED IS AT 230V WITHOUT AIR FILTER IN PLACE.
2. THE CHART IS FOR INFORMATION ONLY. FOR SATISFACTORY OPERATION, EXTERNAL STATIC PRESSURE MUST NOT EXCEED 0.3 IN. WC. CFM VALUES ARE BASED ON 0% TRIM, REFER TO OUTDOOR SPEC SHEET FOR HIGHER/LOWER TRIM VALUES.

**NOTE:** A removable plug connector is provided with the control to make thermostat wire connections. This plug may be removed, wire connections made to the plug, and replaced. It is **STRONGLY** recommended that you do not connect multiple wires into a single terminal. Wire nuts are recommended to ensure one wire is used for each terminal. Failure to do so may result in intermittent operation.

### 11.1 AUXILIARY ALARM SWITCH

The control is equipped with two Auxiliary Alarm terminals, labeled TB4 and TB5 which are typically utilized in series with a condensate switch but could also be used with compatible CO<sub>2</sub> sensors or fire alarms.

The auxiliary alarm switch must be normally closed and open when the alarm occurs. For example, a normally closed condensate switch will open when the base pan's water level reaches a particular level. The control will respond by turning off the blower motor and outdoor unit and displaying the proper fault codes. If the switch is later detected closed for 30 seconds, normal operation resumes and the error message is removed. (The switch is closed as part of the default factory setting.) The error will be maintained in the equipment's fault history. See Figures 4 and 5 for the connection location.

**IMPORTANT NOTE:** If any of the refrigerant leak detection related error codes (A0/AF/A1) are detected at the same time, the action taken when the leak detection error is detected (fan on and relay K4R energized) takes priority.

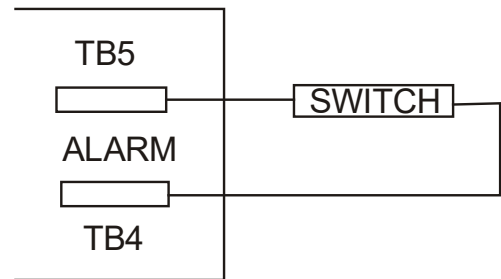
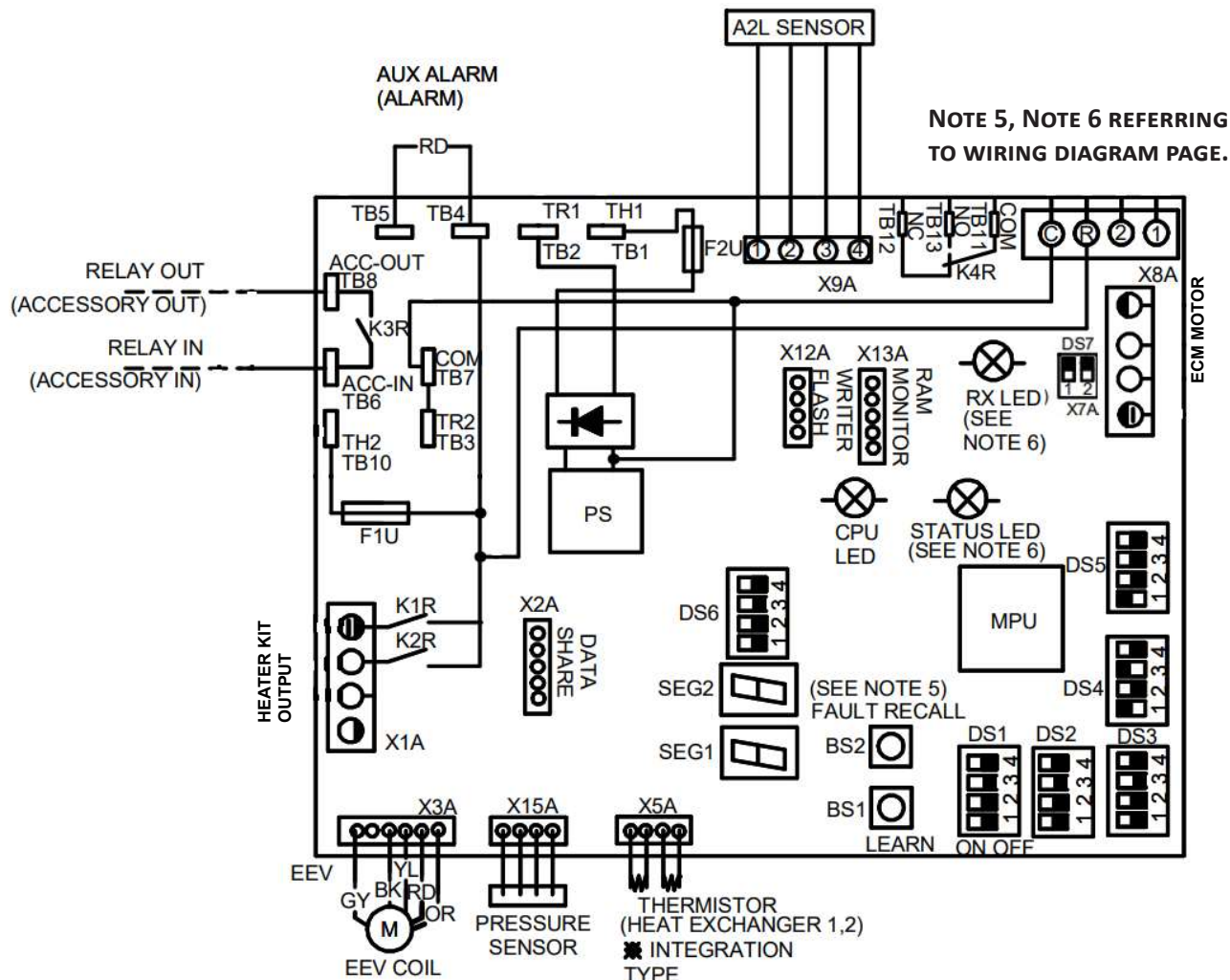


FIGURE 5

### 11.2 CIRCULATOR BLOWER

This air handler is equipped with a variable speed circulator blower. This blower provides several automatically-adjusted blower speeds. For heater kit installation, it is important to set the capacity of the electric heater at two locations of communication thermostat and DIP Switch.



INDOOR UNIT COMMUNICATING BOARD (PCB) - REPRESENTATIVE FIGURE

FIGURE 4

### SETTING WITH COMMUNICATION THERMOSTAT:

For a detailed procedure of thermostat heater kit capacity selection process, please visit the Daikin *One* website at <http://www.daikinone.com> and Goodman GTST Connected Thermostat website at <https://www.goodmanmfg.com/products/controls>.

### SETTING WITH DIP SWITCH:

In the event of loss of communication, emergency mode can be activated. In emergency mode operation, heater kit selection will be driven by the DIP Switch (S9, S10, S11 and S12) selection from the control board. Select the DIP Switch settings in accordance with Tables 7 and 8 (see table 7 for valid settings). Indoor blower airflow (CFM) for a particular heater kit selection can be checked with the STATUS menu of communication thermostat or with the 7-segment display on the control board. (See service manual, section 2-DIGIT 7 SEGMENT DISPLAYS).

**NOTE:** Upon start up in communicating mode the circuit board may display an “Ed” error. This is an indication that the DIP switches on the control board need to be configured in accordance with the Electric Heating Airflow Table. Configuring the DIP switches and resetting power to the unit will clear the error code.



### CAUTION

DO NOT CHANGE ANY OTHER DIP SWITCHES OTHER THAN S9 TO S12. INCORRECT SETTINGS MAY CAUSE ANY ERROR. FOR DEFAULT SETTING, SEE FIGURE 4.

### AWVE MODELS

| SETTING      |                      |
|--------------|----------------------|
| MODELS       | HEATER KIT SELECTION |
| AWVE12SU1300 | NOT RATED            |
| AWVE18SU1300 | NOT RATED            |
| AWVE24SU1300 | NOT RATED            |
| AWVE30LU1300 | 3                    |
| AWVE36LU1300 | 3                    |

TABLE 7

| DIP SWITCH SETTING   |                        |               |                 |                 |                 |
|----------------------|------------------------|---------------|-----------------|-----------------|-----------------|
| FUNCTION             |                        | SWITCH 9 (S9) | SWITCH 10 (S10) | SWITCH 11 (S11) | SWITCH 12 (S12) |
| HEATER KIT SELECTION | NO HEATER KIT          | OFF*          | OFF*            | OFF*            | OFF*            |
|                      | FIRST VALID HEATER KIT | ON            | ON              | ON              | ON              |

DIP SWITCH SETTING

TABLE 8

### 11.3 ACCESSORY CONTACTS

The control is equipped with an Accessory Relay and a pair of ¼ inch accessory terminals which is normally open, labeled ACC-IN [TB6] and ACC-OUT [TB8]. (This is a Dry contact. See accessory contacts graphic).

The Accessory Relay can be configured to close with humidification functionality or to close anytime the blower is running. A closed relay means the two terminals will have continuity between them. (The control does not provide power to these contacts).

This setting can be made in the “3. Equipment setup menu” of the Daikin *One+*.

#### “3. Equipment Set up -> Air handler-> Humidifier relay”

For the setup for humidification functionality, the accessory terminals have 3 operational modes.

##### 1. humidifier-> humidification mode

###### 1-1: off

Contact (relay) never closes

###### 1-2: on

The contact (relay) closes when, [humidification demand: on] and [heating demand: on] and [blower motor; on]

###### 1-3: independent

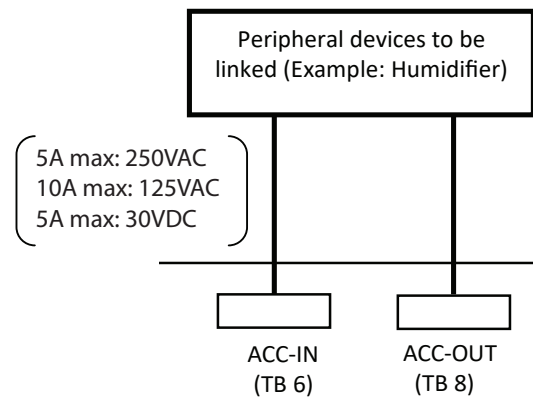
The contact (relay) close when, [humidification demand; on] and [blower motor; on]

##### 2. enable with blower

Contact (relay) closes when, [blower motor; on]

##### 3. non (factory setting)

Contact (relay) never closes



ACCESSORY CONTACTS  
FIGURE 6

## 11.4 LEAK DETECTION OUTPUT (RELAY K4R)



### WARNING

ENSURE THAT ANY ADDITIONAL WIRING FROM THE INDOOR UNIT TO THE FIELD DEVICE IS ROUTED AND PROTECTED FROM DAMAGE AND WEAR, AVOIDING THE FLUE PIPE AND ANY JOINTS THAT MAY NEED BRAZED OR DISCONNECTED FOR SERVICE. ALL FIELD CONNECTED DEVICES (EXCEPT BUILT-IN ELECTRIC HEATER KITS WHICH ARE CONTROLLED BY THE UNIT'S CONTROL BOARD) THAT MAY BECOME A POTENTIAL IGNITION SOURCE MUST BE CONNECTED TO THE LEAK DETECTION OUTPUT (RELAY: K4R) ON THE MODULAR BLOWER CONTROL BOARD. IF A FIELD DEVICE, SUCH AS AN ELECTRONIC AIR CLEANER, IS ALREADY INSTALLED IN THE DUCT WORK AND CANNOT NOT BE INTERLOCKED WITH LEAK DETECTION OUTPUT, IT MUST BE DISABLED OR REMOVED.

The control board is equipped with three Refrigerant Leak Detection 1/4 inch quick connect terminals, labeled TB11, TB12 and TB13 that are used for the control of optional kits. When a refrigerant leak is detected, as indicated by error codes A0, A1 or AF, relay K4R is activated by the control. See the table 9 for the conditions of relay K4R when the error code is issued. See the table 10 for the specification for relay K4R and Figures 4 and 7 for the terminals' location.

| ITEMS<br>CONNECTIONS      | ERROR CODE<br>A0/A1/AF NOT<br>ISSUED | ERROR CODE<br>A0/A1/AF<br>ISSUED |
|---------------------------|--------------------------------------|----------------------------------|
| TB11 (COM) /<br>TB12 (NC) | CLOSED                               | OPEN                             |
| TB11 (COM) /<br>TB13 (NO) | OPEN                                 | CLOSED                           |

RELAY K4R WHEN A0/A1/AF IS ISSUED  
TABLE 9

|                                  |                                                          |
|----------------------------------|----------------------------------------------------------|
| Contact types                    | 1 Form C, Dry contact                                    |
| Contact rating (resistive)       | Form A contact (TB11-13):<br>10 A 125 V AC, 5 A 250 V AC |
|                                  | Form B contact (TB11-12):<br>3 A 125 V AC, 2 A 250 V AC  |
| Max. switching power (resistive) | Form A contact (TB11-13):<br>1,250 VA, 150 W             |
|                                  | Form B contact (TB11-12):<br>500 VA, 30 W                |
| Max. switching current           | Form A contact (TB11-13):<br>10 A (125 V AC)             |
|                                  | Form B contact (TB11-12):<br>3 A (125 V AC)              |

RELAY K4R SPECIFICATIONS  
TABLE 10

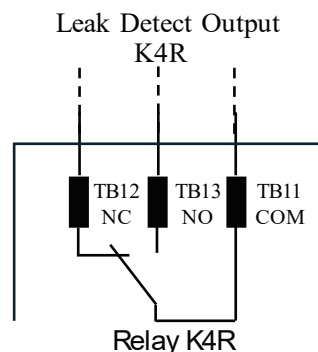


FIGURE 7

## 12 TROUBLESHOOTING

### 12.1 ELECTROSTATIC DISCHARGE (ESD) PRECAUTIONS

**NOTE:** Discharge body's static electricity before touching unit. An electrostatic discharge can adversely affect electrical components.

Use the following precautions during air handler installation and servicing to protect the integrated control module from damage. By putting the air handler, the control, and the person at the same electrostatic potential, these steps will help avoid exposing the integrated control module to electrostatic discharge. This procedure is applicable to both installed and uninstalled (ungrounded) blowers.

1. Disconnect all power to the blower. Do not touch the integrated control module or any wire connected to the control prior to discharging your body's electrostatic charge to ground.
2. Firmly touch a clean, unpainted, metal surface of the air handler blower near the control. Any tools held in a person's hand during grounding will be discharged.
3. Service integrated control module or connecting wiring following the discharge process in step 2. Use caution not to recharge your body with static electricity; (i.e., do not move or shuffle your feet, do not touch ungrounded objects, etc.). If you come in contact with an ungrounded object, repeat step 2 before touching control or wires.
4. Discharge your body to ground before removing a new control from its container. Follow steps 1 through 3 if installing the control on a blower. Return any old or new controls to their containers before touching any ungrounded object.

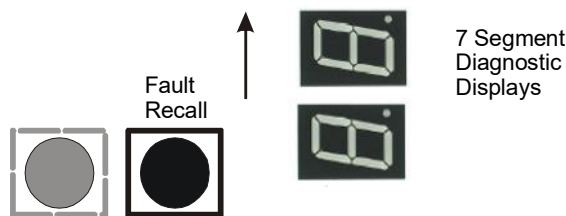


FIGURE 8

**NOTE: 7 SEGMENT DIAGNOSTIC DISPLAYS ARE ROTATED 90-DEGREE COUNTERCLOCKWISE AND ARE VISIBLE THROUGH THE SIGHT GLASS ON THE ACCESS PANEL.**



## 12.2 DIAGNOSTIC CHART

Refer to the *Troubleshooting Chart* at the end of this manual for assistance in determining the source of unit operational problems. The 7 segment LED display will provide any active fault codes.

## 12.3 FAULT RECALL

The integrated control module is equipped with a momentary push-button switch that can be used to display the last six faults on the 7 segment LED display. Follow the sequence to use the feature. The control must be in Standby Mode (no thermostat inputs).

1. Press FAULT RECALL button for 2 to 5 seconds\*, so that 7 segment display shows “- -”.
2. Release FAULT RECALL button in this period, 7 segment display shows the most recent fault.
3. Each time FAULT RECALL button is pressed after that\*\*, 7 segment display outputs next occurred fault.
4. After displaying the series of recent faults, 7 segment displays blink “- -” and goes back to Standby Mode.

To clear the error code history:

1. Press FAULT RECALL button for 10 to 15 seconds\*\*\*, so that 7 segment displays blink “- -”.
2. Release FAULT RECALL button in this period, 7 segment displays show “88” and faults are cleared.

### NOTE:

\* **IF FAULT RECALL BUTTON IS NOT PRESSED LONG ENOUGH (FOR 2 TO 5 SECONDS), CONTROL GOES BACK TO STANDBY MODE.**

**IF THE FAULT RECALL BUTTON IS PRESSED FOR 5 TO 10 SECONDS, CONTROL GOES BACK TO STANDBY MODE.**

\*\* **CONSECUTIVELY REPEATED FAULTS ARE DISPLAYED A MAXIMUM OF THREE TIMES.**

\*\*\* **IF FAULT RECALL BUTTON IS PRESSED FOR LONGER THAN 15 SECONDS, CONTROL GOES BACK TO STANDBY MODE.**

## 13 FULLY COMMUNICATING DAIKIN OR GOODMAN SYSTEM

**NOTE:** For a detailed procedure of thermostat commissioning process, please visit the Daikin *One+* website at <http://www.daikinone.com> and Goodman GTST Connected Thermostat website at <https://www.goodmanmfg.com/products/controls/gtst>.

### 13.1 OVERVIEW

A fully communicating Daikin or Goodman system is a system that includes a compatible air handler and air conditioner or heat pump with a communicating thermostat.

A fully communicating Daikin or Goodman heating/air conditioning system differs from a non-communicating/traditional system in the manner in which the indoor unit, outdoor unit and thermostat interact with one another. In a traditional system, the thermostat sends commands to the indoor and outdoor units via analog 24 VAC signals. It is a one-way communication path. The indoor and outdoor units typically do not return information to the thermostat.

For a fully communicating Daikin or Goodman system, the indoor unit, outdoor unit, and thermostat “communicate” digitally with one another. It is now a two-way communications path. The thermostat still sends commands to the indoor and outdoor units and may also request and receive information from both the indoor and outdoor units. This information may be displayed on the communicating thermostat. The indoor and outdoor units also interact with one another. The outdoor unit may send commands to or request information from the indoor unit. This two-way digital communications between the thermostat and subsystems (indoor/outdoor unit) is the key to unlocking the benefits and features of the fully communicating Daikin or Goodman system.

Two-way digital communications is accomplished using only two wires. The thermostat and air handler controls are powered with 24 VAC. A maximum of 4 wires between the air handler and thermostat is required to operate the system. An inverter equipped outdoor unit does not require 24 VAC. Only the 2 digital communication wires are required between the air handler and inverter unit (pins 1 and 2 on the thermostat connector.)

### 13.2 AIRFLOW CONSIDERATION

Airflow demands are managed differently in a fully communicating system than in a non-communicating wired system. The system operating mode (as determined by the thermostat) determines which unit calculates the system airflow demand. If the indoor unit is responsible for determining the airflow demand, it calculates the demand and sends it to the ECM motor. If the outdoor unit or thermostat is responsible for determining the demand, it calculates the demand and transmits the demand along with a fan request to the indoor unit. The indoor unit then sends the demand to the ECM motor. Table 11 lists the various fully communicating Daikin or Goodman systems, the operating mode, and airflow demand source.

For example, assume the system is a heat pump matched with an air handler. With a call for low stage cooling, the heat pump will calculate the system's low stage cooling airflow demand. The heat pump will then send a fan request along with the low stage cooling airflow demand to the air handler. Once received, the air handler will send the low stage cooling airflow demand to the ECM motor. The ECM motor then delivers the low stage cooling airflow. See the applicable fully communicating Daikin or Goodman System air conditioner or heat pump installation manual for the airflow delivered during cooling or heat pump heating.

In continuous fan mode, the communicating thermostat provides the airflow demand. The communicating thermostat provides 4 continuous fan speeds (25%, 50%, 75% and 100% of maximum airflow). During continuous fan operation, the thermostat sends a fan request along with the continuous fan demand to the air handler. The air handler, in turn, sends the demand to the ECM motor. The ECM motor delivers the requested continuous fan airflow.

### 13.3 THERMOSTAT WIRING

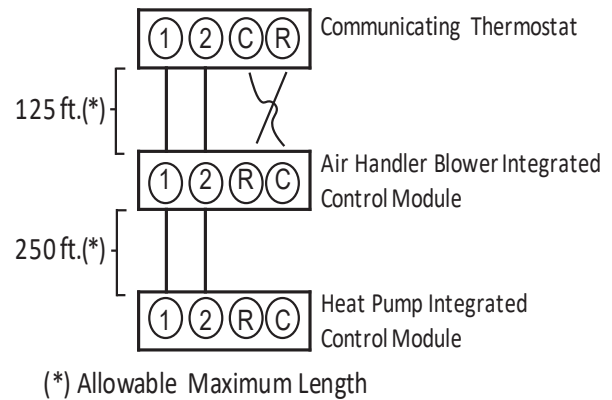
**NOTE:** Refer to section Electrical Connections for 208/230 volt line connections to the air handler.

**NOTE:** It is **STRONGLY** recommended that no more than two wires be connected in a single terminal. If two wires are used in a terminal, it is recommended the same type of wire be used (i.e. Both stranded or solid for secure connection). Failure to do so may result in intermittent operation.

Typical 18 AWG thermostat wire may be used to wire the system components. Two hundred fifty (250) feet is the maximum length of wire between indoor unit and outdoor unit, and one hundred twenty five (125) feet between indoor unit and thermostat.

#### 13.3.1 Two-Wire Outdoor and Four-Wire Indoor Wiring

Typical wiring will consist of two wires between the indoor unit and outdoor unit and four wires between the indoor unit and thermostat. Figure 9 shows the required wires are: data lines, 1 and 2; "R" (24 VAC hot) and "C" (24 VAC common).



**SYSTEM WIRING**  
**FIGURE 9**

### 13.4 NETWORK TROUBLESHOOTING

Occasionally the need to troubleshoot the network may arise. The integrated air handler control has some on-board tools that can be used to troubleshoot the network. These tools are: red communications LED, green receive (Rx) LED, and the learn button. Refer to the Communications Troubleshooting Chart at the end of this manual for error codes, possible causes and corrective actions

- Red communications LED – Indicates the status of the network. The Communications Troubleshooting Chart on the following page indicates the LED status and the corresponding potential problem.
- Green receive LED – Indicates network traffic. The following table indicates the LED status and the corresponding potential problem.
- Learn button – Used to reset the network. Depress the button for approximately 2 seconds to reset the network.

### 13.5 SYSTEM TROUBLESHOOTING

**NOTE:** Refer to the instructions accompanying the Communicating compatible outdoor AC/HP unit for unit specific troubleshooting information. Refer to the *Troubleshooting Chart* at the end of this manual for a listing of possible air handler error codes, possible causes and corrective actions.

## ATTENTION INSTALLER - IMPORTANT NOTICE!

**Please read carefully before installing this unit.**

- Power line terminal #C from Indoor unit must connect to terminal #C on thermostat and power line terminal #R from indoor unit must connect to terminal #R on thermostat. Verify wires are not reversed.  
(Note: The order of the terminals of the indoor unit and the Daikin One+ or Goodman GTST Connected thermostat may be different.)
- Do not attach any wires to the R & C Terminals on the AC/HP, as they are not needed for inverter unit.
- Data line terminal #1 from AC/HP must connect to terminal #1 on indoor unit and thermostat and data line terminal #2 from AC/HP must connect to terminal #2 on indoor unit and thermostat. Verify wires are not reversed.

Servicing: Before servicing identify the Mode of operation of the system by reading the 7-Segment Display on the PCB which can be seen through the round glass view on top access panel and matching the 7-Segment Display with mode of operation in Diagnostic label which is attached on the Blower access panel. After identifying the mode of operation take recommended actions as specified in Table DIAGNOSTIC CODES.

### **13.6.1 Instruction to replace PCB:**

As shown in Figures 10 and 11, take off the blower access panel, disconnect the PCB harness and A2L sensor wire connected to the PCB, detach the PCB from the 4 plastic standoffs, install new PCB on 4 plastic standoffs which is installed on metal bracket, re-connect the PCB harness and A2L sensor wire to the new PCB per wiring instruction as attached to the equipment, reassemble the blower access panel to the unit.

| System                        | System Operating Mode     | Airflow Demand Source                           |
|-------------------------------|---------------------------|-------------------------------------------------|
| Air Conditioner + Air Handler | Cooling                   | Air Conditioner                                 |
|                               | Heating                   | Air Handler                                     |
|                               | Continuous Fan            | Thermostat                                      |
| Heat Pump + Air Handler       | Cooling                   | Heat Pump                                       |
|                               | Heat Pump Heating Only    | Heat Pump                                       |
|                               | HP+ Electric Heat Strips  | Greater than of Heat Pump of Air Handler Demand |
|                               | Electric Heat Strips Only | Air Handler                                     |
|                               | Continuous Fan            | Thermostat                                      |

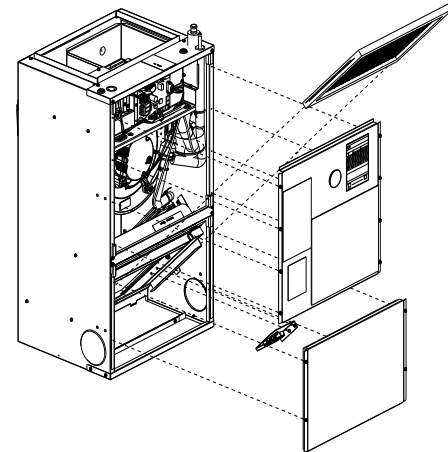
TABLE 11

### **13.6 REFRIGERATION DETECTION SYSTEM (RDS)**

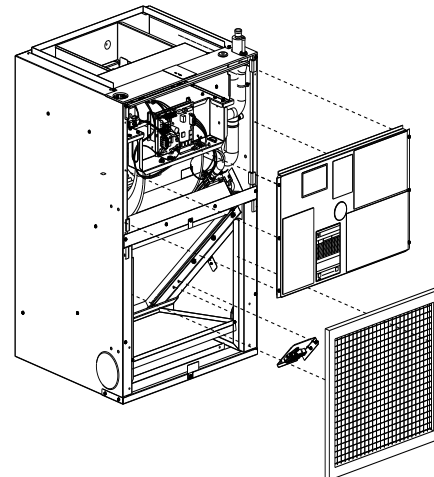
Function: Refrigerant Detection System (RDS) is installed in this equipment to detect any R32 leakage in the coil and take action to mitigate any risk of ignition/fire.

Operation: When there is a leak detected, the RDS shall send signals for the unit to perform these actions:

1. Turn on the blower to circulate air with Mitigation CFM
  2. Switch off electric heater
  3. Error code A0 issued
  4. Leak Detection Output (relay K4R) energized (\*1)
- (\*1) If any accessories that could become a potential ignition source are installed, they must be wired to LEAK DETECTION OUTPUT (RELAY K4R) so that they are deactivated when a leak is detected. Refer to "11.4 LEAK DETECTION OUTPUT (RELAY K4R)" for details.



AWVE 12,18,24  
FIGURE 10



AWVE 30, 36  
FIGURE 11

### 13.6.2 INSTRUCTION TO REPLACE A2L SENSOR:

#### 13.6.2.1 SENSOR DISMOUNTING INSTRUCTIONS:

1. Dismount the front top access panel by unscrewing all the screws.
2. Remove the pin connecting the sensor cable on the PCB as shown in Figures 10 and 11. Remove the filter from the air handler based on the following:
  - A. For AWVE12, 18, 24 units, perform the following:
    - Remove the front bottom panel by unscrewing the 4 screws shown in Figure 10 (In some cases it might have already been removed due to front return setup. Skip this step if that's the case).
    - Take out the filter by slowly sliding it out.
  - B. For AWVE30, 36 units, perform the following:
    - Rotate the sheet metal cover holding the filter.
    - Remove the filter.
3. Dismount the sensor bracket assembly by unscrewing the screws which hold the sensor bracket on the right gutter rail.
4. Remove the R-32 sensor cable by cutting and removing the cable tie without damaging the R-32 sensor cable.
5. Unscrew both the screws which hold the R-32 sensor on the sensor bracket.

#### 13.6.2.2 SENSOR MOUNTING INSTRUCTIONS:

1. Place the sensor on the sensor bracket such that the two holes on the sensor and sensor bracket align with each other.
2. Use two M0211216 screws to fasten the sensor on the sensor bracket using a torque drill using torque values of 15-17 in-lbs.
3. Mount the cable tie on the bottom left portion of the sensor bracket.
4. Place the R-32 sensor wire as shown in Figures 10 and 11 and use the cable tie to secure and tie the R-32 sensor wire. Cut the edge of the cable tie.
5. Mount the sensor bracket assembly to the right gutter rail using two M0211216 screws.
6. Mount the filter back on the air handler based on following:
  - A. For AWVE12, 18, 24 units, perform the following:
    - Place the filter by sliding it in slowly through the filter holder.
    - Mount the front bottom panel by screwing the 4 screws shown in Figure 10 (In some cases it might have already been removed due to front return setup. Skip this step if that's the case).
  - B. For AWVE30, 36 units, perform the following:
    - Place the filter as shown in the Figure 11.
    - Rotate the sheet metal cover to hold the filter in place.

**IMPORTANT: THE R-32 SENSOR SHALL ONLY BE REPLACED WITH SENSORS SPECIFIED BY THE APPLIANCE MANUFACTURING. (SENSOR REPLACEMENT PART CAN BE DETERMINED THROUGH THE HOMEOWNER SUPPORT LINE PROVIDED EARLIER IN THIS DOCUMENT.)**

|                                                                                    |                                           |
|------------------------------------------------------------------------------------|-------------------------------------------|
|  | service indicator; read technical manual  |
|   | operator's manual; operating instructions |
|   | warning; low Flammable materials          |
|  | UN GHS flame symbol                       |

**MARKING SYMBOL  
TABLE 12**

### 13.7 MITIGATION AIRFLOW

**Serial Plate Information:** The installer must write the maximum possible refrigerant charge and the minimum room area on serial plate. Use Table 13 to determine the maximum possible refrigerant charge and minimum (air conditioned) room area based on the total system refrigerant charge. If the installation is not at sea level consult section 13.7.1 to determine how to adjust the minimum (air conditioned) room area. Use a durable marking instrument to write the values on the serial plate.

The installers are responsible for verifying that actuation of mitigation procedures are operational with the minimum required airflow for mitigation.

| TOTAL SYSTEM<br>REFRIGERANT<br>CHARGE | MAXIMUM<br>POSSIBLE<br>REFRIGERANT<br>CHARGE | MAXIMUM<br>POSSIBLE<br>REFRIGERANT<br>CHARGE | MINIMUM AIR<br>CONDITIONED<br>ROOM AREA | MINIMUM AIR<br>CONDITIONED<br>ROOM AREA | MINIMUM<br>REQUIRED<br>MITIGATION<br>AIRFLOW | MINIMUM<br>REQUIRED<br>MITIGATION<br>AIRFLOW |
|---------------------------------------|----------------------------------------------|----------------------------------------------|-----------------------------------------|-----------------------------------------|----------------------------------------------|----------------------------------------------|
| (oz)                                  | (oz)                                         | (KG)                                         | (FT2)                                   | (M2)                                    | (CFM)                                        | (M3/HR)                                      |
| 31~50                                 | 50                                           | 1.42                                         | 46                                      | 4.22                                    | 83                                           | 140                                          |
| 51~100                                | 100                                          | 2.84                                         | 91                                      | 8.44                                    | 165                                          | 279                                          |
| 101~150                               | 150                                          | 4.26                                         | 137                                     | 12.66                                   | 247                                          | 418                                          |
| 151~200                               | 200                                          | 5.67                                         | 182                                     | 16.85                                   | 328                                          | 556                                          |
| 201~250                               | 250                                          | 7.09                                         | 227                                     | 21.07                                   | 410                                          | 696                                          |
| 251~300                               | 300                                          | 8.51                                         | 273                                     | 25.29                                   | 492                                          | 835                                          |
| 301~350                               | 350                                          | 9.93                                         | 318                                     | 29.51                                   | 574                                          | 974                                          |
| 351~400                               | 400                                          | 11.34                                        | 363                                     | 33.69                                   | 655                                          | 1112                                         |
| 401~450                               | 450                                          | 12.76                                        | 409                                     | 37.91                                   | 737                                          | 1251                                         |
| 451~500                               | 500                                          | 14.18                                        | 454                                     | 42.13                                   | 819                                          | 1391                                         |
| 501~550                               | 550                                          | 15.60                                        | 499                                     | 46.35                                   | 901                                          | 1530                                         |
| 551~600                               | 600                                          | 17.01                                        | 545                                     | 50.54                                   | 982                                          | 1668                                         |

A2L MITIGATION TABLE

TABLE 13

NOTE: FOR MITIGATION CFM, WHEN ONE OF THE ERROR CODES A0/AF/A1 THAT IS RELATED TO REFRIGERANT LEAK DETECTION SYSTEM IS ISSUED, THE BLOWER IS ACTIVATED WITH THE MITIGATION CFM ABOVE THE QMIN SHOWN IN TABLE 13.

### 13.7.1 ALTITUDE ADJUSTMENT FACTOR TO CALCULATE MINIMUM ROOM AREA

For altitudes other than at sea level, adjust the minimum room area specified on the Serial Plate by the corresponding altitude adjustment factor shown below. This table is provided as a reference. Adjusted room area (Amin adj) is the product of the minimum room area specified in the serial plate and the adjustment factor AF, as shown in below formula.

Amin adj = Amin (as per table 13) \* AF

| HEIGHT IN METERS | HEIGHT IN FEET | ALTITUDE<br>ADJUSTMENT FACTOR<br>(AF) |
|------------------|----------------|---------------------------------------|
| AT SEA LEVEL     | AT SEA LEVEL   | 1.00                                  |
| 1~200            | 1~660          | 1.02                                  |
| 200~400          | 660~1320       | 1.03                                  |
| 400~600          | 1320~1970      | 1.05                                  |
| 600~800          | 1970~2630      | 1.07                                  |
| 800~1000         | 2630~3290      | 1.09                                  |
| 1000~1200        | 3290~3940      | 1.11                                  |
| 1200~1400        | 3940~4600      | 1.13                                  |
| 1400~1600        | 4600~5250      | 1.15                                  |
| 1600~1800        | 5250~5910      | 1.17                                  |
| 1800~2000        | 5910~6570      | 1.19                                  |
| 2000~2200        | 6570~7220      | 1.21                                  |
| 2200~2400        | 7220~7880      | 1.24                                  |
| 2400~2600        | 7880~8540      | 1.26                                  |
| 2600~2800        | 8540~9190      | 1.29                                  |
| 2800~3000        | 9190~9850      | 1.31                                  |
| 3000~3200        | 9850~10500     | 1.34                                  |

A\_MIN ALTITUDE CORRECTION FACTOR

TABLE 14

NOTE: THE ALTITUDE ADJUSTMENT FACTOR (AF) FOR ANY ALTITUDES BELOW SEA LEVEL WILL BE 1.00.

### 13.8 REFRIGERANT LEAK TEST



#### WARNING

**DURING REFRIGERANT LEAK TEST, THE BLOWER ACTIVATES, AND LEAK DETECTION OUTPUT (RELAY K4R) IS ENERGIZED.**

Conduct Refrigerant Leak Test to check that the refrigerant leak detection system works when optional accessory kits are installed. To start Refrigerant Leak Test, go to "Refrigerant Leak Detection" menu under "4 System Optimization" on the thermostat's commissioning menu. Switch "Refrigerant leak test" setting to "Run test" to start the refrigerant leak test. When the unit goes into Refrigerant Leak Test, the message will be displayed on the thermostat to inform that it's in testing mode. After confirming these actions, change the setting for "Refrigerant leak test" to "Stop" to finish the test.

**NOTE: THE REFRIGERANT LEAK TEST IS AUTOMATICALLY TURNED OFF AFTER 1 HOUR, EVEN IF YOU DO NOT SET IT TO "STOP"**

### 13.9 DETECTION OF REFRIGERANT

Under no circumstances shall potential sources of ignition



be used in the searching for or detection of refrigerant leaks. A halide torch (or any other detector using a naked flame) shall not be used. The following leak detection methods are deemed acceptable for all refrigerant systems. Electronic leak detectors may be used to detect refrigerant leaks but, in the case of FLAMMABLE REFRIGERANTS, the sensitivity may not be adequate, or may need recalibration. (Detection equipment shall be calibrated in a refrigerant-free area.) Ensure that the detector is not a potential source of ignition and is suitable for the refrigerant used. Leak detection equipment shall be set at a percentage of the LFL of the refrigerant and shall be calibrated to the refrigerant employed, and the appropriate percentage of gas (25 % maximum) is confirmed. Leak detection fluids are also suitable for use with most refrigerants but the use of detergents containing chlorine shall be avoided as the chlorine may react with the refrigerant and corrode the copper pipe-work.

If a leak is suspected, all naked flames shall be removed/ extinguished. If a leakage of refrigerant is found which requires brazing, all of the refrigerant shall be recovered from the system, or isolated (by means of shut off valves) in a part of the system remote from the leak.

### **13.10 REMOVAL AND EVACUATION**

When breaking into the refrigerant circuit to make repairs - or for any other purpose - conventional procedures shall be used. However, for flammable refrigerants it is important that best practice be followed, since flammability is a consideration. The following procedure shall be adhered to. The refrigerant charge shall be recovered into the correct recovery cylinders if venting is not allowed by local and national codes. For appliances containing flammable refrigerants, the system shall be purged with oxygen free nitrogen to render the appliance safe for flammable refrigerants. This process might need to be repeated several times. Compressed air or oxygen shall not be used for purging refrigerant systems.

### **13.11 REFRIGERANT RECOVERY**

When removing refrigerant from a system, either for servicing or decommissioning, all refrigerants must be removed safely. When transferring refrigerant into cylinders, ensure that only appropriate refrigerant recovery cylinders are employed. Ensure that the correct number of cylinders for holding the total system charge is available. All cylinders to be used are designated for the recovered refrigerant and labeled for that refrigerant (i.e. special cylinders for the recovery of refrigerant). Cylinders shall be complete with pressure-relief valve and associated shut-off valves in good working order. Empty recovery cylinders are evacuated and, if possible, cooled before recovery occurs.

# 14 TROUBLESHOOTING

| Error Code | PCB LED Display                     | Description                                                                                                                                                                        | Possible Causes                                                                                                                                                                                                                        | Corrective Actions                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EE         | No display (EE display is EMG mode) | <ul style="list-style-type: none"> <li>No power supply to ID blower / no 24 volt power to PCB</li> <li>Blown fuse or circuit breaker</li> <li>PCB has an internal fault</li> </ul> | <ul style="list-style-type: none"> <li>Manual disconnect switch OFF</li> <li>No power supply to ID blower / no 24 volt power to PCB</li> <li>Blown fuse or faulty circuit breaker</li> <li>Control board has internal fault</li> </ul> | <ul style="list-style-type: none"> <li>Assure 208/230 volt and 24 volt power to blower and control board.</li> <li>Check fuse F2U on control board</li> <li>Check for possible short in 208/230 volt and 24 volt circuits. Repair as necessary.</li> <li>Replace the control board.</li> </ul>                                                                                                                |
| Eb         | E_Eb                                | Selecting "no heater kit" and receiving electric heat demand                                                                                                                       | <ul style="list-style-type: none"> <li>No heater kit selected</li> </ul>                                                                                                                                                               | <ul style="list-style-type: none"> <li>Select the valid heater kit on thermostat</li> <li>Valid dip switch selection (heater kit selection out of range of the unit configuration)</li> </ul>                                                                                                                                                                                                                 |
| Ed         | E_Ed                                | Heater Kit dip switches not set properly                                                                                                                                           | <ul style="list-style-type: none"> <li>Invalid heater kit selected</li> </ul>                                                                                                                                                          | <ul style="list-style-type: none"> <li>Set correct dip switches</li> </ul>                                                                                                                                                                                                                                                                                                                                    |
| E5         | E_E5                                | Fuse Open                                                                                                                                                                          | <ul style="list-style-type: none"> <li>Fuse (F1U) is blown</li> <li>Connector TB10 is open</li> </ul>                                                                                                                                  | <ul style="list-style-type: none"> <li>Replace fuse</li> <li>Check wiring to AUX alarm, heater kit, communication connection.</li> <li>Replace the control board</li> </ul> <p><i>After recovering the system, E5 will still be displayed on the indoor PCB (about 30 seconds). BLOWN FUSE will still be displayed on the thermostat within 45 seconds. They will be cleared automatically.</i></p>           |
| EF         | E_EF                                | Auxiliary Switch Open                                                                                                                                                              | <ul style="list-style-type: none"> <li>High water level in the evaporation coil</li> <li>The connected alarm device is activated</li> <li>Auxiliary Alarm terminals (TB4, TB5) are open</li> </ul>                                     | <ul style="list-style-type: none"> <li>Check water level in drain pan</li> <li>Check alarm device.</li> <li>Close Auxiliary terminals TB4 and TB5 if not used</li> </ul> <p><i>After recovering the system, EF will still be displayed on the indoor PCB (about 30 seconds). AUXILIARY CONTACTS OPEN will still be displayed on the thermostat within 45 seconds. They will be cleared automatically.</i></p> |
| d0         | E_d0                                | Data not on Network                                                                                                                                                                | <ul style="list-style-type: none"> <li>No shared data on the network</li> </ul>                                                                                                                                                        | <ul style="list-style-type: none"> <li>Populate shared data set using memory card.</li> </ul>                                                                                                                                                                                                                                                                                                                 |
| d1         | E_d1                                | Invalid Data on Network                                                                                                                                                            | <ul style="list-style-type: none"> <li>Wrong shared data on the network</li> </ul>                                                                                                                                                     | <ul style="list-style-type: none"> <li>Populate shared data set using memory card.</li> </ul>                                                                                                                                                                                                                                                                                                                 |
| d4         | E_d4                                | Invalid Memory Card Data                                                                                                                                                           | <ul style="list-style-type: none"> <li>Wrong memory card data</li> </ul>                                                                                                                                                               | <ul style="list-style-type: none"> <li>Replace circuit board</li> <li>Rewrite data using the correct memory card</li> </ul>                                                                                                                                                                                                                                                                                   |
| b0         | E_b0                                | Blower Motor not running                                                                                                                                                           | <ul style="list-style-type: none"> <li>Fan/motor obstruction</li> <li>Power interruption (low voltage)</li> <li>Incorrect / loose wiring</li> </ul>                                                                                    | <ul style="list-style-type: none"> <li>Check for obstruction on the fan/motor</li> <li>Verify the input voltage at the motor</li> <li>Check wiring or tighten wiring connections if needed</li> <li>Replace circuit board or motor</li> </ul>                                                                                                                                                                 |
| b1         | E_b1                                | Blower Motor Communication error                                                                                                                                                   | <ul style="list-style-type: none"> <li>Incorrect / loose wiring</li> <li>Power interruption (low voltage)</li> </ul>                                                                                                                   | <ul style="list-style-type: none"> <li>Check wiring or tighten wiring connections if needed</li> <li>Verify the input voltage at the motor</li> <li>Replace circuit board or motor</li> </ul>                                                                                                                                                                                                                 |
| b2         | E_b2                                | Blower Motor HP Mismatch                                                                                                                                                           | <ul style="list-style-type: none"> <li>Incorrect size motor</li> <li>Invalid shared data</li> </ul>                                                                                                                                    | <ul style="list-style-type: none"> <li>Correct motor installation</li> <li>Populate shared data set using memory card.</li> </ul>                                                                                                                                                                                                                                                                             |
| b3         | E_b3                                | Blower Motor operating in Power, Temp or Speed Limiting conditions                                                                                                                 | <ul style="list-style-type: none"> <li>Fan/motor obstruction or blocked filters</li> <li>Power interruption (low voltage)</li> <li>Incorrect wiring</li> <li>Blockage in the airflow (ductwork) or ductwork undersized</li> </ul>      | <ul style="list-style-type: none"> <li>Check for obstruction on the fan/motor/ductwork, clean filters</li> <li>Verify the input voltage at the motor</li> <li>Check wiring</li> <li>Replace motor</li> </ul>                                                                                                                                                                                                  |

# TROUBLESHOOTING

| Error Code | PCB LED Display | Description                                                                   | Possible Causes                                                                                                                                                                                                                                                                                                     | Corrective Actions                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------|-----------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| b9         | E_b9            | Low Indoor Airflow<br>(without Electric Heat mode)                            | <ul style="list-style-type: none"> <li>Fan/motor obstruction or blocked filters</li> <li>Restrictive ductwork or ductwork undersized</li> <li>Wiring disconnected</li> <li>Wrong outdoor/indoor combination</li> <li>ID motor failure</li> </ul>                                                                    | <ul style="list-style-type: none"> <li>Check for obstruction on the fan/motor</li> <li>Check ductwork/filter for blockage, clean filters</li> <li>Remove obstruction. Verify all registers are fully open</li> <li>Check the connections and the rotation of the motor</li> <li>Verify the input voltage at the motor</li> <li>Verify ductwork is appropriately sized for system. Resize/replace ductwork if needed</li> <li>Replace motor</li> </ul>                                    |
| 9b         | E_9b            | Low Indoor Airflow<br>(with Electric Heat mode)                               | <ul style="list-style-type: none"> <li>Fan/motor obstruction or blocked filters</li> <li>Restrictive ductwork or ductwork undersized</li> <li>ID motor failure</li> <li>Wiring disconnected</li> </ul>                                                                                                              | <ul style="list-style-type: none"> <li>Check for obstruction on the fan/motor</li> <li>Check ductwork/filter for blockage, clean filters</li> <li>Remove obstruction. Verify all registers are fully open</li> <li>Check the connections and the rotation of the motor</li> <li>Verify the input voltage at the motor</li> <li>Verify ductwork is appropriately sized for system. Resize/replace ductwork if needed</li> <li>Replace motor</li> </ul>                                    |
| 70         | E_70            | EEV disconnection detected                                                    | <ul style="list-style-type: none"> <li>Indoor EEV coil not connected</li> <li>Incorrect wiring to EEV</li> </ul>                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>Check Indoor EEV coil connection (PCB and junction connector)</li> <li>Replace EEV coil</li> <li>Check the resistance value of EEV coil (refer service manual)</li> <li>Replace the control board</li> </ul>                                                                                                                                                                                                                                      |
| 73         | E_73            | Liquid side thermistor abnormality                                            | <ul style="list-style-type: none"> <li>Open (or) short circuit of the liquid thermistor (X5A)</li> <li>Liquid thermistor reading incorrect or values outside the normal range</li> </ul>                                                                                                                            | <ul style="list-style-type: none"> <li>Check the connection to liquid thermistor (PCB and junction connector)</li> <li>Check the resistance value of the thermistor (refer service manual)</li> <li>Replace thermistor</li> <li>Replace the control board</li> </ul>                                                                                                                                                                                                                     |
| 74         | E_74            | Gas side thermistor abnormality                                               | <ul style="list-style-type: none"> <li>Open (or) short circuit of the gas thermistor (X5A)</li> <li>Gas thermistor reading incorrect or values outside the normal range</li> </ul>                                                                                                                                  | <ul style="list-style-type: none"> <li>Check the connection to gas thermistor (PCB and junction connector)</li> <li>Check the resistance value of the thermistor (refer service manual)</li> <li>Replace thermistor</li> <li>Replace the control board</li> </ul>                                                                                                                                                                                                                        |
| 75         | E_75            | Pressure sensor abnormality                                                   | <ul style="list-style-type: none"> <li>Open (or) short circuit of the Pressure sensor (X15A)</li> <li>Pressure sensor reading incorrect or values outside the normal range</li> </ul>                                                                                                                               | <ul style="list-style-type: none"> <li>Check the connection to pressure sensor (PCB and junction connector)</li> <li>Check the output voltage of the pressure sensor (refer service manual)</li> <li>Replace pressure sensor</li> <li>Replace the control board</li> </ul>                                                                                                                                                                                                               |
| 77         | E_77            | Indoor Unit - Thermostat communication error<br>(start-up & during operation) | <ul style="list-style-type: none"> <li>Incorrect wiring between ID unit and thermostat.<br/><i>The system may have the communication error without error code 77 on the indoor PCB. Follow section 17.5 SYSTEM TROUBLESHOOTING</i></li> <li>Thermostat failure</li> <li>Power interruption (low voltage)</li> </ul> | <ul style="list-style-type: none"> <li>Check for thermostat and indoor unit wiring</li> <li>Verify the input voltage at the ID unit and thermostat<br/><i>After recovering the system with power supply, TSTAT ID NO COM will continue to be displayed on the thermostat within 2 minutes. The error code will be cleared automatically.</i></li> <li>Replace control board or thermostat</li> <li>Press "LEARN" button on PCB for more than 5 seconds to reestablish network</li> </ul> |
| A0         | E_A0            | Refrigerant Leak Alarm                                                        | <ul style="list-style-type: none"> <li>Refrigerant Leak</li> <li>A2L sensor failure</li> </ul>                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Ventilate the room before conducting any actions</li> <li>Check coil leakage</li> <li>Remove refrigerant from system and replace coil</li> </ul>                                                                                                                                                                                                                                                                                                  |
| AF         | E_AF            | A2L Sensor Communication Error                                                | <ul style="list-style-type: none"> <li>A2L sensor not connected</li> <li>Incorrect wiring of A2L sensor</li> </ul>                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>Check the connection of A2L sensor</li> <li>Replace A2L sensor</li> </ul>                                                                                                                                                                                                                                                                                                                                                                         |
| A1         | E_A1            | A2L Sensor Internal Alarm                                                     | <ul style="list-style-type: none"> <li>A2L sensor failure</li> </ul>                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Replace A2L sensor</li> <li>Replace control board</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                      |

## 15 NETWORK TROUBLESHOOTING

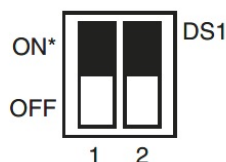
If a network communication error code has occurred, use the following steps to help troubleshoot the system. (For network communication error codes, refer to the table below and the tables of error codes for outdoor unit and indoor unit.)

After any wiring changes have been made or the dip switches of DS1 on the outdoor control board and/or DS7 on the indoor unit control board have been changed, apply power to the system and see if the error codes have cleared.

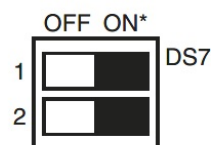
1. Confirm low voltage wiring is correct per installation instructions. Check for miswiring. (i.e. Terminal 1 and 2 is reversed.)

NOTE: A removable plug connector is provided with the control to make thermostat wire connections. This plug may be removed, wire connections made to the plug, and replaced. It is strongly recommended that you do not connect more than two wires into a single terminal in the field because there is a risk of the wires becoming loose, which may result in intermittent operation.

2. Check wires for damage. (i.e. Broken wire at terminal, broken inside wire nuts or damaged cable between units.)
3. Perform continuity check on wires to make sure cable is OK. Replace the cable if necessary.
4. Change both dip switch 1 and 2 of DS1 on the outdoor unit control board. In case the indoor unit control board has DS7, refer to the following Table 15. Try the combinations of dip switches for DS1 and DS7 on the outdoor and indoor unit control board, respectively, one by one. Check that any combinations clear the communication error. These dip switches change the termination resistance value of the communication circuit.



Outdoor unit DS1



Indoor unit DS7

(\*)Default factory setting.

**Dip Switch Setting Combinations**

| Setting Combinations | DS1 on the outdoor unit control board | DS7 on the indoor unit control board |
|----------------------|---------------------------------------|--------------------------------------|
| 1*                   | (both) ON                             | (both) ON                            |
| 2                    | (both) OFF                            | (both) ON                            |
| 3                    | (both) ON                             | (both) OFF                           |
| 4                    | (both) OFF                            | (both) OFF                           |

(\*) Default factory setting

**TABLE 15**

The integrated control module has some onboard tools that can be used to troubleshoot the network. Refer to Table 16. These tools are: red communications LED, green receive (Rx) LED, and the learn button.

- Red communications LED – Indicates the status of the network. The table below indicates the LED status and the corresponding potential problem.
- Green receive LED – Indicates network traffic. The table below indicates the LED status and the corresponding potential problem.
- LEARN button – Used to reset the network. Press the button for approximately 5 seconds to reset the network.

| LED COLOR                    | LED Status     | Indication                       | Probable Causes                                                                                                                                                                                                                                                                                                 | Corrective Actions                                                                                                                                                                                                                                                                                         |
|------------------------------|----------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Red Communications LED (H2P) | OFF            | Normal condition                 | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                                                                                                                                                                                   |
|                              | 1 Flash        | Communication failure            | <ul style="list-style-type: none"> <li>• Unknown packet is received</li> <li>• Communications failure</li> </ul>                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>• Depress learn button</li> <li>• Verify wiring connection</li> </ul>                                                                                                                                                                                               |
|                              | 2 Flash        | Out-of-box reset                 | <ul style="list-style-type: none"> <li>• Control power up</li> <li>• Learn button depressed</li> </ul>                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                                                                                                                                                                                   |
| Green Receive LED (H3P)      | OFF            | No power<br>Communications error | <ul style="list-style-type: none"> <li>• No power to unit</li> <li>• Open fuse</li> <li>• Communication error</li> </ul>                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Check circuits breakers and fuses; Reset/Replace if needed</li> <li>• Reset network by depressing learn button</li> <li>• Check communication wires (terminal 1/terminal 2 wires); Replace if needed</li> <li>• Check for shorts in low voltage wiring</li> </ul> |
|                              | 1 Steady Flash | No network found                 | <ul style="list-style-type: none"> <li>• Broken/disconnected communication wire(s)</li> <li>• Unit is installed as a legacy/traditional system</li> </ul>                                                                                                                                                       | <ul style="list-style-type: none"> <li>• Check communication wires (terminal 1/terminal 2 wires); Replace if needed</li> <li>• Check installation type (legacy/traditional or communicating)</li> </ul>                                                                                                    |
|                              | Rapid Flashing | Normal network traffic           | <ul style="list-style-type: none"> <li>• Control is "taking" on network as expected</li> </ul>                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• None</li> </ul>                                                                                                                                                                                                                                                   |
|                              | On Solid       | Terminal 1/Terminal 2 miss-wire  | <ul style="list-style-type: none"> <li>• Terminal 1 and Terminal 2 wires reversed at indoor unit, thermostat, or outdoor unit</li> <li>• Short between terminal 1 and terminal 2 wires</li> <li>• Short between terminal 1 or terminal 2 two wires and terminal C (24VAC) or terminal R (24VAC, COM)</li> </ul> | <ul style="list-style-type: none"> <li>• Check communication wires (terminal 1/terminal 2 wires); Replace if needed</li> </ul>                                                                                                                                                                             |

**TABLE 16**

## 16 DIAGNOSTIC CODES

| 7 SEGMENT LED<br>DISPLAY                                                                                                                          | DESCRIPTION OF CONDITION                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| (NO DISPLAY)                                                                                                                                      | INTERNAL CONTROL FAULT / NO POWER                                                      |
| On                                                                                                                                                | STANDBY, WAITING FOR INPUTS                                                            |
| EE                                                                                                                                                | EMERGENCY MODE RUNNING                                                                 |
| Eb                                                                                                                                                | NO HEATER KIT INSTALLED - SYSTEM CALLING FOR AUXILIARY HEAT (MINOR ERROR CODE)         |
| Ed                                                                                                                                                | HEATER KIT DIP SWITCHES NOT SET PROPERLY                                               |
| E5                                                                                                                                                | FUSE OPEN                                                                              |
| EF                                                                                                                                                | AUXILIARY SWITCH OPEN                                                                  |
| d0                                                                                                                                                | DATA NOT ON NETWORK                                                                    |
| d1                                                                                                                                                | INVALID DATA ON NETWORK                                                                |
| d4                                                                                                                                                | INVALID MEMORY CARD DATA                                                               |
| b0                                                                                                                                                | BLOWER MOTOR NOT RUNNING                                                               |
| b1                                                                                                                                                | BLOWER MOTOR COMMUNICATION ERROR                                                       |
| b2                                                                                                                                                | BLOWER MOTOR HP (HORSE POWER) MISMATCH                                                 |
| b3                                                                                                                                                | BLOWER MOTOR OPERATING IN POWER, TEMPERATURE, OR SPEED LIMIT                           |
| b9                                                                                                                                                | LOW INDOOR AIRFLOW (MINOR ERROR CODE) (WITHOUT ELECTRICAL HEATER MODE)                 |
| 9b                                                                                                                                                | LOW INDOOR AIRFLOW<br>(MAJOR ERROR CODE) (ELECTRICAL HEATER MODE ONLY)                 |
| 70                                                                                                                                                | EEV OPEN CIRCUIT                                                                       |
| 73                                                                                                                                                | LIQUID TEMPERATURE ABNORMALITY / THERMISTOR OPEN CIRCUIT                               |
| 74                                                                                                                                                | GAS TEMPERATURE ABNORMALITY / THERMISTOR OPEN CIRCUIT                                  |
| 75                                                                                                                                                | PRESSURE ABNORMALITY / PRESSURE SENSOR OPEN CIRCUIT                                    |
| 77                                                                                                                                                | INDOOR UNIT - THERMOSTAT COMMUNICATION ERROR<br>(STARTUP OPERATION & DURING OPERATION) |
| Hu                                                                                                                                                | HUMIDIFICATION DEMAND (RUNNING WITHOUT HEATING)                                        |
| A0                                                                                                                                                | A2L REFRIGERANT LEAKAGE ALARM                                                          |
| AF                                                                                                                                                | A2L SENSOR COMM ALARM                                                                  |
| A1                                                                                                                                                | A2L SENSOR INTERNAL ALARM                                                              |
| <p>7SEG LED WILL FLASH AIRFLOW RATE</p>  <p>0140A01620-A</p> |                                                                                        |



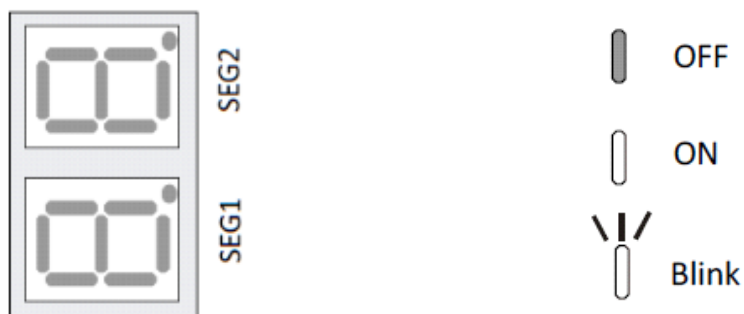
## 17 SETTING THE MODE DISPLAY

### MODE DISPLAY INTRODUCTION

A 2-digit display is provided on the printed circuit board (PCB) as a backup tool to the thermostat for accessing error codes and erasing error code history of the indoor unit. Follow the information provided in this section to learn how to use the mode display

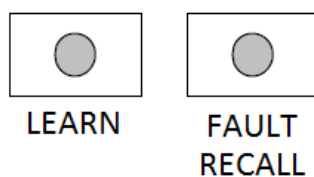
### DISPLAY

The display consists of 2 digits.



### DISPLAY BUTTON LAYOUT

The display buttons shown can be used to navigate and select items:



*Example of button layout is shown above*

### FAULT CODE HISTORY NAVIGATION

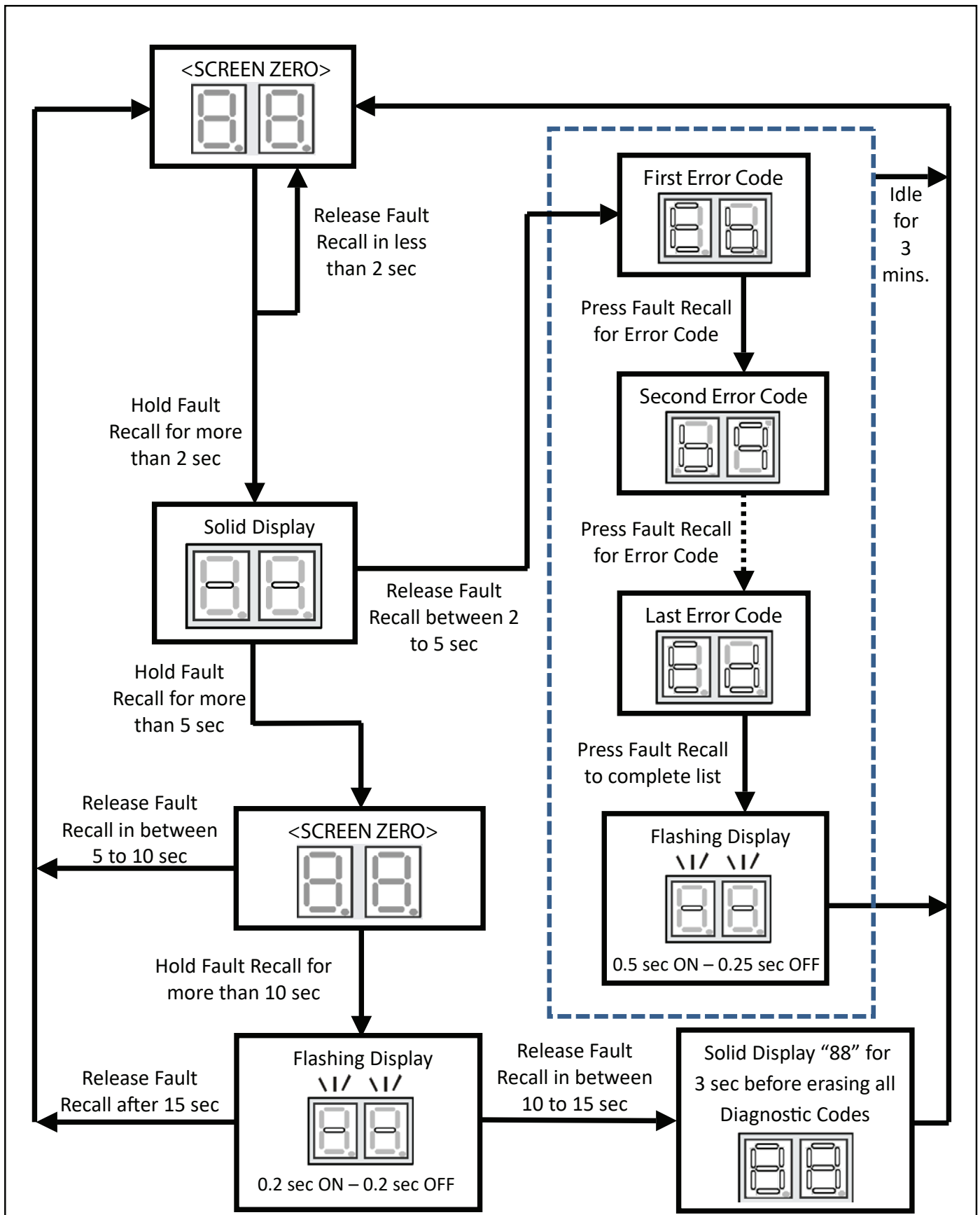
This mode will allow the user to see the six most recent system faults. Please follow the flow chart to navigate to error codes from screen zero.

For a list of the fault codes, please see the TROUBLESHOOTING tables in this document.

It is also possible to erase all the diagnostics codes from this menu.

## SETTING THE MODE DISPLAY

**NOTE:** 7 SEGMENT DIAGNOSTIC DISPLAYS ARE ROTATED 90-DEGREE COUNTERCLOCKWISE AND ARE VISIBLE THROUGH THE SIGHT GLASS ON THE ACCESS PANEL



# 18 WIRING DIAGRAM

**NOTE:** THESE INSTRUCTIONS ARE SPECIFICALLY FOR AWVE12, AWVE18, AWVE24 MODELS. DO NOT ATTEMPT TO APPLY THESE DIAGRAMS FOR ANY OTHER MODELS.

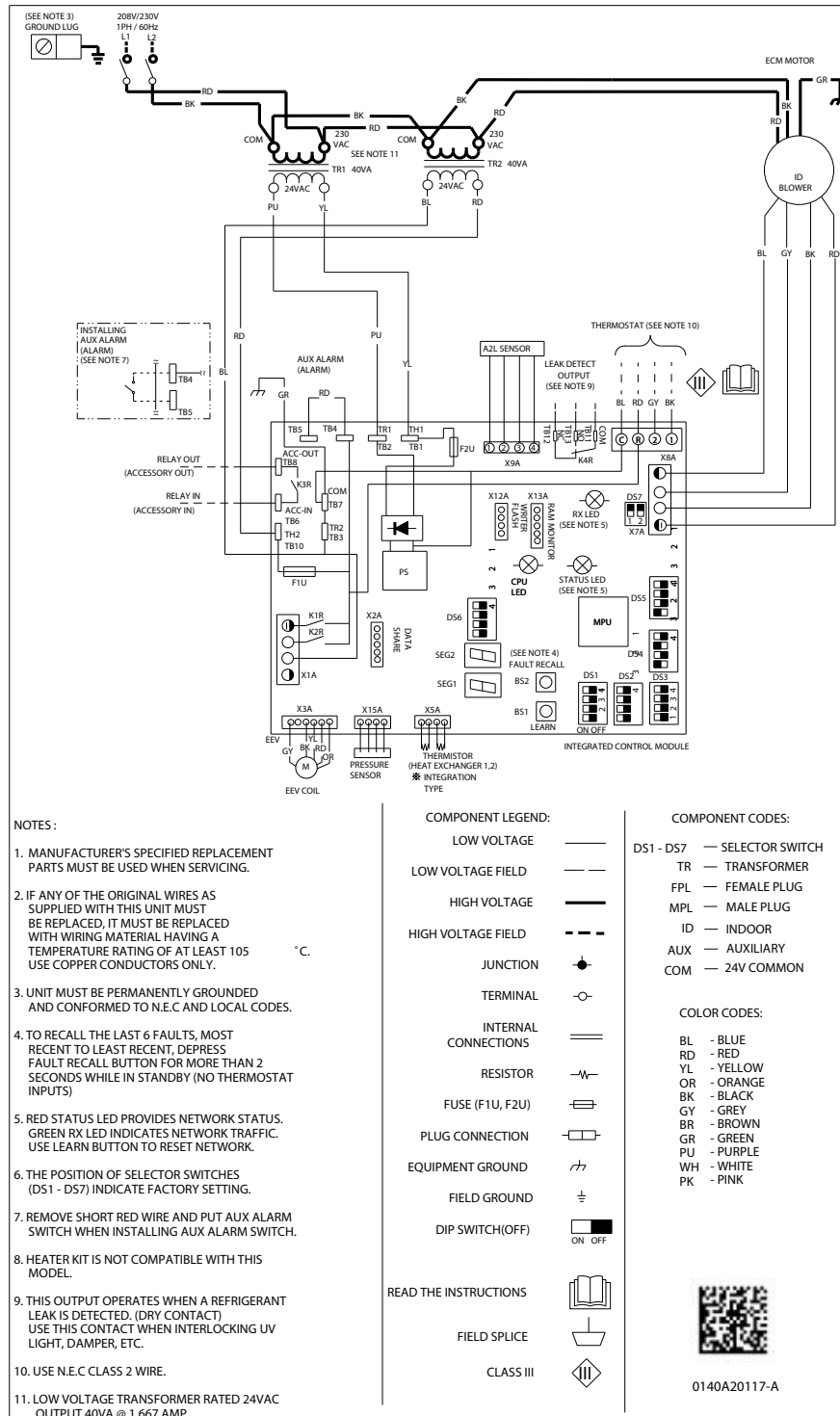


## WARNING

**HIGH VOLTAGE! DISCONNECT ALL POWER BEFORE SERVICING.**

**MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.**





Wiring is subject to change. Always refer to the wiring diagram on the unit for the most up-to-date wiring.

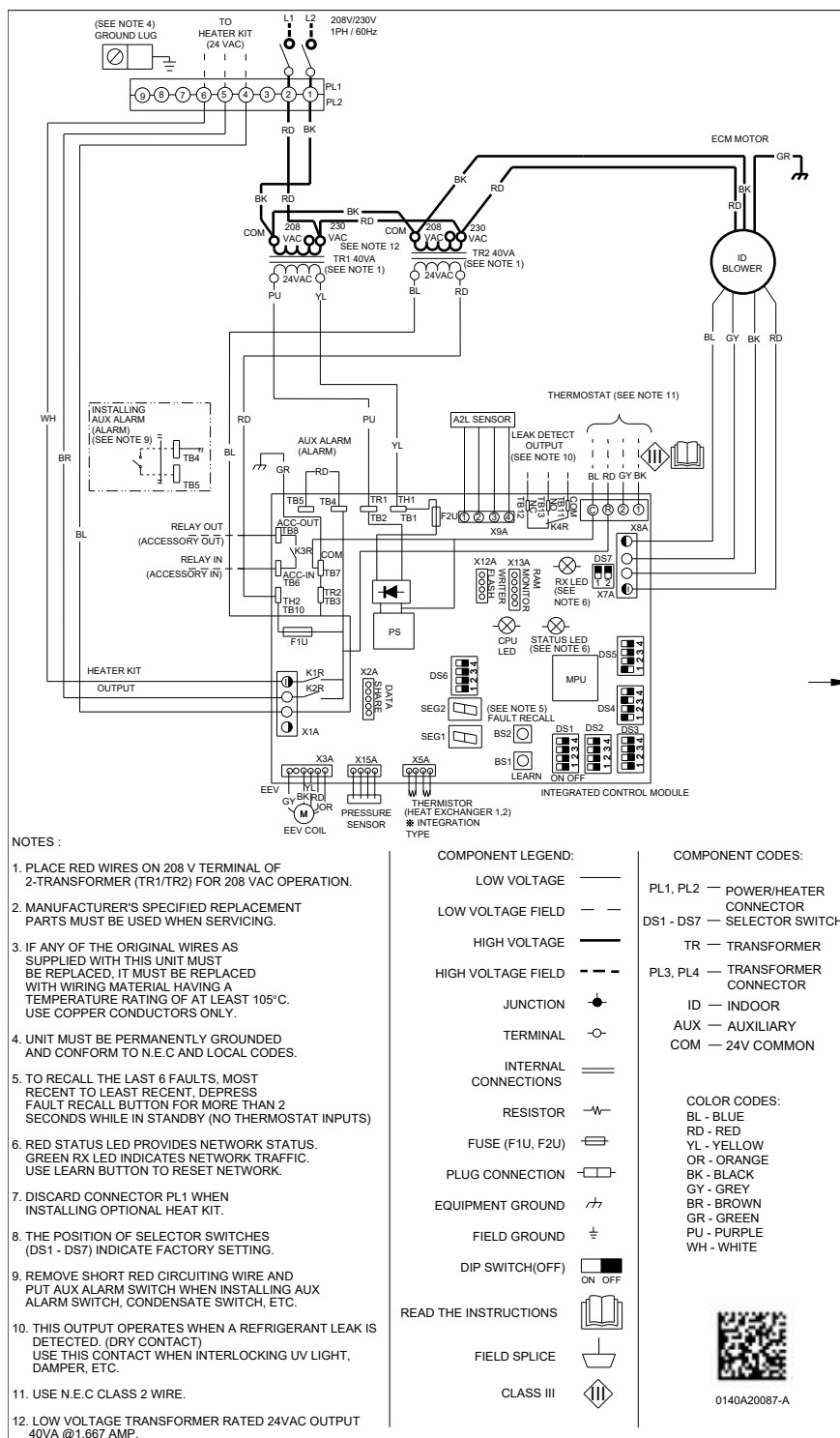
# WIRING DIAGRAM

**NOTE:** THESE INSTRUCTIONS ARE SPECIFICALLY FOR AWVE30, AWVE36 MODELS. DO NOT ATTEMPT TO APPLY THESE DIAGRAMS FOR ANY OTHER MODELS.



**WARNING**

**HIGH VOLTAGE! DISCONNECT ALL POWER BEFORE SERVICING.**  
**MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO**  
**SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.**



Wiring is subject to change. Always refer to the wiring diagram on the unit for the most up-to-date wiring.

## SPLIT SYSTEMS

### AIR CONDITIONING AND HEAT PUMP HOMEOWNER'S ROUTINE MAINTENANCE RECOMMENDATIONS

We strongly recommend a biannual maintenance checkup be performed before the heating and cooling seasons begin by a qualified servicer.

#### REPLACE OR CLEAN FILTER

**IMPORTANT NOTE: NEVER OPERATE UNIT WITHOUT A FILTER INSTALLED AS DUST AND LINT WILL BUILD UP ON INTERNAL PARTS RESULTING IN LOSS OF EFFICIENCY, EQUIPMENT DAMAGE AND POSSIBLE FIRE.**



#### WARNING

##### HIGH VOLTAGE!

**DISCONNECT ALL POWER BEFORE SERVICING OR INSTALLING THIS UNIT. MULTIPLE POWER SOURCES MAY BE PRESENT. FAILURE TO DO SO MAY CAUSE PROPERTY DAMAGE, PERSONAL INJURY OR DEATH.**



An indoor air filter must be used with your comfort system. A properly maintained filter will keep the indoor coil of your comfort system clean. A dirty coil could cause poor operation and/or severe equipment damage.

Your air filter or filters could be located in your furnace, in a blower unit, or in "filter grilles" in your ceiling or walls. The installer of your air conditioner or heat pump can tell you where your filter(s) are, and how to clean or replace them.

Check your filter(s) at least once a month. When they are dirty, replace or clean as required. Disposable type filters should be replaced. Reusable type filters may be cleaned.

You may want to ask your dealer about high efficiency filters. High efficiency filters are available in both electronic and non-electronic types. These filters can do a better job of catching small airborne particles.

#### COMPRESSOR

The compressor motor is hermetically sealed and does not require additional oiling.

#### MOTORS

Indoor and outdoor fan motors are permanently lubricated and do not require additional oiling.

#### ALUMINUM INDOOR COIL CLEANING (QUALIFIED SERVICER ONLY)

R-32 sensor must be removed from the unit before applying coil cleaners to avoid damage and contamination. This unit is equipped with an aluminum tube evaporator coil. The safest way to clean the evaporator coil is to simply flush the coil with water. This cleaning practice remains as the recommended cleaning method for both copper tube and aluminum tube residential cooling coils.

An alternate cleaning method is to use one of the products listed in the technical publication TP-109\* (shipped in the literature bag with the unit) to clean the coils. The cleaners listed are the only agents deemed safe and approved for use to clean round tube aluminum coils. TP-109 is available on the web site in Partner Link > Service Toolkit.

**NOTE: ENSURE COILS ARE RINSED WELL AFTER USE OF ANY CHEMICAL CLEANERS.**

#### CLEAN OUTSIDE COIL (QUALIFIED SERVICER ONLY)

Air must be able to flow through the outdoor unit of your comfort system. Do not construct a fence near the unit or build a deck or patio over the unit without first discussing your plans with your dealer or other qualified servicer. Restricted airflow could lead to poor operation and/or severe equipment damage.

Likewise, it is important to keep the outdoor coil clean. Dirt, leaves, or debris could also restrict the airflow. If cleaning of the outdoor coil becomes necessary, hire a qualified servicer. Inexperienced people could easily puncture the tubing in the coil. Even a small hole in the tubing could eventually cause a large loss of refrigerant. Loss of refrigerant can cause poor operation and/or severe equipment damage.

Do not use a condensing unit cover to "protect" the outdoor unit during the winter, unless you first discuss it with your dealer. Any cover used must include "breathable" fabric to avoid moisture buildup.

#### Before Calling your Servicer



#### CAUTION

**TO AVOID THE RISK OF EQUIPMENT DAMAGE OR FIRE, INSTALL THE SAME AMPERAGE BREAKER OR FUSE AS YOU ARE REPLACING. IF THE CIRCUIT BREAKER OR FUSE SHOULD OPEN AGAIN WITHIN THIRTY DAYS, CONTACT A QUALIFIED SERVICER TO CORRECT THE PROBLEM. IF YOU REPEATEDLY RESET THE BREAKER OR REPLACE THE FUSE WITHOUT HAVING THE PROBLEM CORRECTED, YOU RUN THE RISK OF SEVERE EQUIPMENT DAMAGE.**

- Check the thermostat to confirm that it is properly set.
- Wait 15 minutes. Some devices in the outdoor unit or in programmable thermostats will prevent compressor operation for awhile, and then reset automatically. Also, some power companies will install devices which shut off air conditioners for several minutes on hot days. If you wait several minutes, the unit may begin operation on its own.
- Check the electrical panel for tripped circuit breakers or open fuses. Reset the circuit breakers or replace fuses as necessary.
- Check the disconnect switch near the indoor furnace or blower to confirm that it is closed.
- Check for obstructions on the outdoor unit. Confirm that it has not been covered on the sides or the top. Remove any obstruction that can be safely removed. If the unit is covered with dirt or debris, call a qualified servicer to clean it.
- Check for blockage of the indoor air inlets and outlets. Confirm that they are open and have not been blocked by objects (rugs, curtains or furniture).
- Check the filter. If it is dirty, clean or replace it.
- Listen for any unusual noise(s), other than normal operating noise, that might be coming from the outdoor unit. If you hear unusual noise(s) coming from the unit, call a qualified servicer.



## 19 START-UP CHECKLIST

| Air Handler / Coil                                                                                      |         |               |             |
|---------------------------------------------------------------------------------------------------------|---------|---------------|-------------|
|                                                                                                         |         | Model Number  | _____       |
|                                                                                                         |         | Serial Number | _____       |
| <b>ELECTRICAL</b>                                                                                       |         |               |             |
| Line Voltage (Measure L1 and L2 Voltage)                                                                | L1 - L2 | _____         |             |
| Secondary Voltage (Measure Transformer Output Voltage)                                                  | R - C   | _____         |             |
| Blower Amps                                                                                             |         | _____         |             |
| Heat Strip 1 - Amps                                                                                     |         | _____         |             |
| Heat Strip 2 - Amps                                                                                     |         | _____         |             |
| <b>BLOWER EXTERNAL STATIC PRESSURE</b>                                                                  |         |               |             |
| Return Air Static Pressure                                                                              |         | _____         | IN. W.C.    |
| Supply Air Static Pressure                                                                              |         | _____         | IN. W.C.    |
| Total External Static Pressure (Ignoring +/- from the reading above, add total here)                    |         | _____         | IN. W.C.    |
| <b>TEMPERATURES</b>                                                                                     |         |               |             |
| Return Air Temperature (Dry bulb / Wet bulb)                                                            | _____   | DB °F         | _____ WB °F |
| Cooling Supply Air Temperature (Dry bulb / Wet bulb)                                                    | _____   | DB °F         | _____ WB °F |
| Heating Supply Air Temperature                                                                          | _____   | DB °F         |             |
| Temperature Rise                                                                                        | _____   | DB °F         |             |
| Delta T (Difference between Supply and Return Temperatures)                                             | _____   | DB °F         |             |
| Air Handler / Coil - (Inverter Matched)                                                                 |         |               |             |
| <b>INVERTER AH / COIL ONLY</b>                                                                          |         |               |             |
| Check EEV and EEV wiring is secure (no adjustment required)                                             |         | _____         |             |
| Additional Checks                                                                                       |         |               |             |
| Check wire routings for any rubbing                                                                     |         | _____         |             |
| Check product for proper draining                                                                       |         | _____         |             |
| Check screw tightness on blower wheel                                                                   |         | _____         |             |
| Check factory wiring and wire connections                                                               |         | _____         |             |
| Check product for proper clearances as noted by installation instructions                               |         | _____         |             |
| °F to °C formula: (°F - 32) divided by 1.8 = °C      °C to °F formula: (°C multiplied by 1.8) + 32 = °F |         |               |             |

**THIS PAGE IS INTENTIONALLY LEFT BLANK**

**THIS PAGE IS INTENTIONALLY LEFT BLANK**

#### **CUSTOMER FEEDBACK**

Daikin Comfort Technologies is very interested in all product comments.

Please fill out the feedback form on the following link:

<https://daikincomfort.com/contact-us>

You can also scan the QR code on the right to be directed to the feedback page.



NOTE: SPECIFICATIONS AND PERFORMANCE DATA LISTED HEREIN ARE SUBJECT TO CHANGE WITHOUT NOTICE.

Visit our website at [www.daikincomfort.com](http://www.daikincomfort.com) or [www.goodmanmfg.com](http://www.goodmanmfg.com) for information on:

- |                     |                                    |
|---------------------|------------------------------------|
| • Products          | • Parts                            |
| • Warranties        | • Contractor Programs and Training |
| • Customer Services | • Financing Options                |